Making User Interfaces Accessible for Older Adults

A small usability study of text size, navigation, contrast and loading feedback on a smartphone

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Abstract

This study examined how text size, navigation, contrast and loading feedback can help adults aged 65 and older with little digital experience to book an appointment and find information on a website with their smartphone, independently and with as little frustration and uncertainty as possible. We built a working prototype of a fictional physiotherapy practice in which each of these four features could be switched on or off. Five participants aged 70 and older completed 19 tasks on a smartphone: 11 appointment bookings and 8 information searches. In each task, one feature was switched off while the other three stayed on. We analysed screen recordings, observation notes and an open-ended questionnaire. Four of the ten assessed bookings and six of the eight information tasks were successful. Hidden navigation caused the clearest problems: three of four participants searched for the next step, and one could not find it. Smaller text coincided with more errors when choosing a service and a time, although this may partly be a learning effect. Lower contrast made reading harder for two participants but did not lead to failed tasks, and we observed no effect of missing loading feedback. Several problems were unrelated to the four features, such as a prominent booking button that distracted participants during information tasks. Because the sample was small and each feature was mostly tested with one type of task, the results are preliminary. We have made the prototype and materials public so that others can repeat the study.

Keywords: older adults, accessibility, usability, mobile web, text size, navigation, contrast, loading feedback

Preface

This is the English version of our *profielwerkstuk*, the research project that every student in the final years of Dutch pre-university education (VWO) carries out. The original was written in Dutch. For this version we translated the full report, adapted it for readers outside our school and made it public, together with the prototype and materials, so that others can repeat or extend the study.

We are Yannis Raijmakers and Billie Mastenbroek, students at Scholengemeenschap Were Di in Valkenswaard. Older people we know often struggle with websites and apps. For example, they cannot manage to make a hospital appointment through a website. We think that is not acceptable, and that UI and UX design should take older people more into account. That is why we chose this topic.

In this project we combined a literature review, the design and construction of a working website, and a user study with older adults.

We would like to thank the five participants who took the time to test our website and told us honestly what they found difficult. We also thank our supervisor, [TO FILL IN: name], for their guidance.

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1. Introduction

In this study we investigate how the design of a website can help older adults to book an appointment and find information on their phone without help. To do so, we built a working prototype of a fictional physiotherapy practice and tested it with five participants aged 70 and older.

Chapter 2 describes the problem and reviews the literature on older adults, digital interfaces and accessible design, and translates it into design choices for the prototype. Chapter 3 presents the main research question, the sub-questions and our hypothesis. Chapter 4 describes how we carried out the study. Chapter 5 presents the results, which we use in Chapter 6 to answer the sub-questions and the main question. Chapter 7 discusses the limitations of the study and what could have been done better, and Chapter 8 looks back on the process. The references and appendices follow at the end, with the tasks, the questionnaire and an overview of every task that was carried out.

The prototype was built in Dutch for Dutch participants. Where we mention text from the website, such as a button label, we give the Dutch original in italics followed by an English translation, for example the *Verder* (Next) button. Quotes from participants were translated from Dutch by us.

2. Theoretical framework

2.1 Problem statement

Digital user interfaces are becoming essential to everyday life, and older people are increasingly falling behind. For younger generations, who grew up with smartphones and apps, using them feels natural. For older adults (65+) this is often not the case. Digital products are usually designed with younger users as the norm, and as a result older adults risk exclusion, dependence on others and a loss of autonomy.

The Dutch population is ageing quickly. By 2040, about a quarter of the population is expected to be 65 or older (Centraal Bureau voor de Statistiek, 2024). At the same time, local services are going digital at a high pace: municipal portals, DigiD (the Dutch government login), healthcare apps and public transport systems. Many older people in towns and villages, such as those in the Dutch province of North Brabant where our school is located, rely on family members or libraries to carry out simple digital tasks. The interfaces of these systems are designed for efficiency rather than accessibility, which makes them unnecessarily hard to use for older adults.

According to Google's own research on Material 3 Expressive (Google, n.d.-a), "expressive design", the deliberate use of colour, shape, size, motion and containment, allowed older participants to find key interface elements as quickly as younger participants, while an age difference is normally found in usability tests. Yet many apps that older adults use every day do not apply such principles: buttons are small, colour contrast is low and there is no clear visual hierarchy. This makes these apps harder to use than they need to be. Our main research question is therefore:

How can text size, navigation, contrast and loading feedback contribute to a website on which adults aged 65 and older with little digital experience can book an appointment and find information on their phone independently and with as little frustration and uncertainty as possible?

2.2 Literature review

2.2.1 Human perception and cognitive processing

Before an interface can be designed well, it is necessary to understand how people process visual information. A screen always contains more information than a user can notice at once. Buttons, text, images, colours and empty space all compete for attention. This section therefore covers the basics: how visual attention works, how visual information is organised on a screen, and how much a person can process without becoming overloaded. This knowledge is the starting point for the following sections, which look at how this process changes in older adults and what that means for good interface design.

Visual attention

Attention determines which information on a screen is actually noticed and which is ignored. Selection is necessary because the brain cannot process everything at once: more information always comes in than can be processed. Moore and Zirnsak (2017) call this selection one of the most basic functions of the brain, especially in humans, for whom vision is the most important sense.

In the brain, two routes determine where someone looks. Huang et al. (2023) describe one route related to goal-directed looking: you consciously search for something, such as a button with a particular name. The other route works on the basis of salience: a bright colour or a sudden movement attracts attention, even if you were not looking for it. Both routes work together. One allows you to search for what you need; the other makes sure that important things still stand out when you are not actively looking for them.

This distinction matters for interface design. A screen must support targeted searching (users can find what they are looking for) and must also make important buttons or messages stand out enough to be seen without searching. Zhu and Yang (2023) studied this with students looking at the interface of a learning platform while their eye movements were tracked. They found clear differences in how long and how often students looked at certain parts of the screen. They concluded that colour, text and typography have the greatest influence on where attention goes, while the overall layout of the screen plays a more supporting role. Good use of colour and text can therefore steer what users look at first.

Finally, attention can fade over time. Huang et al. (2023) explain that holding attention for a longer period (sustained attention) underlies other forms of attention, such as searching for something. For interfaces that people use for longer periods, such as a learning platform or an app that is opened every day, it is therefore not only important that individual elements stand out. The screen as a whole must also be arranged so that attention is maintained, without unnecessary distractions.

Visual hierarchy

Whereas visual attention determines what we see, visual hierarchy determines the order in which we process the information on a screen. Without a clear structure an interface is visually “flat” and every element competes equally for priority. Eye-tracking studies show that people rarely read a screen from left to right like a book. Instead, they scan an interface in recurring patterns, such as the F-pattern (Nielsen, 2006; see Figure 8). Elements at the top left, larger elements and elements with high contrast against the background are noticed first. By varying these properties deliberately, the designer determines the path of the eye.

However, not everyone recognises such a hierarchy in the same way. In a questionnaire among 200 internet users, Qing et al. (2024) found that users’ backgrounds are related to the visual elements they pay attention to. Basic stimuli such as a bright colour work for everyone, but how quickly someone understands the structure of a page can differ. It is therefore plausible that users with little digital experience have more difficulty with subtle cues, such as small differences in font weight or complex menus.

The Gestalt principles also play a major role in how we perceive this structure. These are psychological rules that describe how the human brain automatically creates patterns and groups from separate elements. According to the principle of proximity, we immediately see objects that are close together as one group. The principle of similarity causes us to attribute the same function to elements with the same colour or shape, such as a row of buttons in a menu.

A clear hierarchy is therefore essential for good interface design: the most important action button must stand out immediately from the other elements. When this visual structure is missing, the user has to search constantly. This unnecessary searching puts extra pressure on the brain, which leads to the mental load discussed below.

Mental load

Mental load, or cognitive load, is the amount of mental effort that working memory needs at a given moment to understand and use an interface. Human working memory can hold only a limited amount of information at a time. John Sweller, the founder of Cognitive Load Theory, explains that total mental load consists of several components (Sweller, 1988). An important component is extraneous cognitive load, which is caused entirely by the way information is presented (Sweller, 2010). When an interface is unclear, the user spends valuable mental effort on understanding the screen instead of on the task itself (Sweller, 1988).

Arana-De las Casas et al. (2023) tested these limits of memory by having participants operate interfaces while simultaneously performing demanding memory tasks (dual n-back tasks). Their results confirm that when an interface forces users to decipher an unclear menu structure, remember instructions and correct input errors at the same time, working memory becomes overloaded. Once this saturation point is reached, efficiency collapses: users quickly start making mistakes and frustration increases sharply.

To reduce extraneous load, designers can make use of psychological laws. Hick’s law states that the time needed to make a decision increases logarithmically with the number of choices (Hick, 1952). By limiting the number of buttons or menu options, the reaction time and choice stress of a user can be reduced considerably. Grouping information into logical categories also helps, because it allows the brain to filter options more quickly.

When an interface falls short in this respect, the invisible mental load leaves physical traces. Shahnabati et al. (2025) show that physiological eye-tracking measures, such as pupil size and how often the eyes jump back and forth, are reliable indicators of this overload. When an interface contains too much text or unclear icons, the eyes respond with restless searching behaviour and the pupils become measurably larger (pupillometry).

This physiological effect shows that a poorly organised screen can cause stress that is noticeable in the body. Even when a strong visual hierarchy guides the eye well, an excess of simultaneous information can still overload memory. To keep mental load low, the design must minimise the number of thinking steps: by using familiar patterns, removing unnecessary choices and always showing clearly what the system is doing.

2.2.2 Changes with ageing

The previous section described how people process visual information, how attention is directed and how an interface can cause mental load. These processes change as people age. Ageing affects both cognitive functions and the visual system, so that older adults face different challenges than younger users when they use digital interfaces. This section describes these changes and what they mean for the design of usable interfaces.

Cognitive changes

As people age, several cognitive functions involved in using digital interfaces change. In particular, processing speed, working memory and the ability to sustain attention decline on average. This does not mean that older adults are less able to think or learn, but certain tasks can take more time and effort.

Stamate et al. (2024) describe how ageing affects several cognitive processes that are important when using technology. Older adults, for example, more often have difficulty processing large amounts of information at once. As a result, a busy screen can lead to cognitive overload sooner than in younger users.

These changes are also visible in visual search tasks. Wu et al. (2022) found that older participants needed more time to find the right icons on a screen and were more often distracted by similar symbols (see also Affordance in 2.2.4). Searching for functions within an interface therefore takes more cognitive effort as age increases.

This is important for interface design. When users have more difficulty processing information, a clear structure becomes even more important. Simple navigation, recognisable icons and a limited amount of information per screen can help to reduce cognitive load.

In addition, Zhou et al. (2022) found that older users perform better when interface elements are clearly arranged and have distinct visual features. A logical layout helps them spend less mental energy on understanding the interface and more on the actual task.

Visual changes (contrast, size, speed)

Besides cognitive changes, the visual system also changes with age. The eye becomes less sensitive to contrast, focusing takes more effort and visual information is processed more slowly. These changes have direct consequences for the use of digital screens.

Fu et al. (2024) found that visual impairments in older adults are associated with poorer cognitive performance. When text or buttons are hard to see, this costs extra effort, so that visual and cognitive problems can reinforce each other.

One of the most common changes is a decline in contrast sensitivity. Older adults therefore have more difficulty perceiving subtle colour differences. For this reason, the accessibility guidelines of the W3C set a minimum contrast between text and background (World Wide Web Consortium [W3C], 2018); these standards are discussed in 2.2.4.

The size of interface elements also matters. Yeh (2020) studied how older adults operate medical devices with touchscreens. The results show that larger font sizes and well-placed buttons lead to fewer errors and faster use. Small text and buttons placed close together make use considerably more difficult.

For designers, this means that default settings are not always sufficient. Larger text, clear contrast and larger touch targets can make an interface more accessible for older users (Yeh, 2020).

Why standard interfaces fall short

Many modern interfaces are designed for a broad user group. In practice, however, these designs are often based on younger, digitally experienced users. As a result, standard interfaces do not always match the needs of older adults.

Lam (2024) argues that digital accessibility remains an obstacle for people with age-related changes, and that guidelines written for people with disabilities do not always give older adults a good experience. Small text, complex navigation structures and overcrowded screens can be significant obstacles for older adults.

Amouzadeh et al. (2025) likewise conclude in their systematic review that many mobile applications do not follow age-friendly design principles. Complicated menus, hidden functions and unclear icons in particular cause recurring problems. A familiar example is TikTok, where a lot of information is shown on the screen and text is often placed over a video that is already cognitively demanding.

Moreover, older adults are a very diverse user group. Jiang et al. (2024) stress that age-friendly interfaces must take different levels of digital experience into account. What seems obvious to an experienced user can be confusing to a less experienced one.

One example of design work aimed at the needs of older users comes from Google: Material 3 Expressive. This design system uses large touch targets, clear contrast and a consistent visual hierarchy.



Figure 1. Example screens designed according to Google's Material 3 Expressive guidelines. Source: Google (n.d.-b).

Material 3 is partly inspired by design principles that were used earlier to make digital interfaces easier to understand and more intuitive. One of these design styles is skeuomorphism. The idea behind skeuomorphism is that digital elements resemble objects that users already know from the real world, so that people can apply existing knowledge to a digital environment. A button, for example, is given shadows and depth so that it looks like a physical button that can be pressed. Icons are also often designed as recognisable objects, such as an envelope for e-mail or a waste bin for deleting files.

Figure 2 compares early skeuomorphic icons with their modern counterparts.

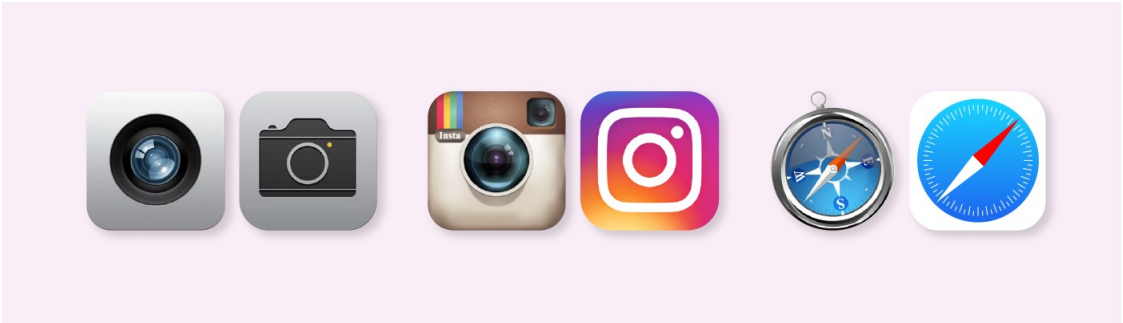


Figure 2. Comparison of early skeuomorphic icons (left) and their modern, simplified versions (right).

Although skeuomorphism was not originally developed specifically for older adults, many of its principles address problems that are more common among older users. Because digital functions resemble familiar everyday objects, users need to spend less time interpreting a

screen. This lowers cognitive load and makes interfaces more accessible for people who find it harder to process new or complex information.

Material 3 adopts these principles in a more modern form. Instead of fully imitating physical objects, the system uses subtle shadows, differences in elevation and animations to make clear which elements are interactive. This keeps the interface uncluttered, while users still receive visual cues that relate to objects and interactions from the real world.

When interfaces do not meet these needs well enough, this can lead to digital exclusion. Mubarak and Suomi (2022) describe this as the growing “grey digital divide”: a situation in which older adults have less access to digital services because existing technology does not match their abilities.

To design accessible interfaces, it is therefore necessary to take the cognitive and visual changes discussed in this section into account. Only then can digital products remain usable for users of all ages.

2.2.3 Emotion and user experience

The previous sections focused mainly on the cognitive and visual processing of interfaces, but the emotional side of interaction design matters as well. Using a digital system is not a purely mechanical process; it constantly evokes emotional reactions. For older users, who more often feel insecure in an increasingly digital society, these emotions can weigh heavily. An interface that evokes positive emotions can lower barriers and support learning, while negative emotions such as frustration and stress can lead to avoidance and digital exclusion. This section describes the relationship between emotion and user experience (UX), how specific feelings arise during interaction, and what role emotional design plays in practice.

Positive emotions: satisfaction, trust and engagement

Positive emotions play an important role in older adults’ acceptance and use of technology. When an interface responds intuitively and the user successfully completes a task, such as sending a message or logging in to a healthcare portal without help, this creates a sense of satisfaction and autonomy. Such moments of success can build the digital self-confidence of older users.

According to Marques et al. (2021), a good user experience goes beyond measuring task functionality or efficiency. Understanding subtle emotional shifts during use is essential for understanding why some designs succeed and others do not. When users experience a sense of control, brief moments of satisfaction can gradually turn into deeper trust in the technology. For older adults, this long-term trust is an important condition for continuing to use an application.

Thoughtful emotional design also contributes to user engagement. Gao and Huang (2022) emphasise that emotional design within human–computer interaction (HCI) allows products to build a meaningful psychological connection with the user. When visual elements, colours and forms of interaction are matched to the emotional needs of the target group, the interface feels

less like a cold, technical obstacle and more like a supportive tool. For older adults, this means that an interface that conveys calm, warmth and clarity can increase their motivation to keep using the technology.

Negative emotions: frustration, stress and confusion

Negative emotions have the opposite effect. When an interface is unclearly structured, buttons do not respond as expected or error messages are cryptic, users become confused. In older adults this confusion can quickly turn into stress and frustration, especially when they blame themselves for the error ("I must have pressed the wrong button again") rather than the design of the interface.

These negative emotions are not only a subjective experience; they can also be observed during usability tests. In their research on emotion recognition, Drungilas et al. (2024) show that software usability is directly related to measurable expressions of frustration, stress and confusion. When a system fails to provide clear feedback or a logical hierarchy, emotion recognition systems register facial expressions of stress and discomfort in the participants.

This stress reaction takes up working memory that would otherwise be available for the task. On top of the extraneous load described in 2.2.1, emotional stress makes a task even harder, and task performance can drop sharply. For an older user, the step from frustration to giving up can be small: after several negative experiences in a row, avoidance can develop, which further widens the grey digital divide.

The role of emotion in interface design

To understand how visual interface elements translate into positive or negative emotions, the three-level model of emotional design by cognitive psychologist Donald Norman (2004) is useful. Norman states that people process products and interfaces at three levels: the visceral, the behavioural and the reflective level.

The visceral level. This is the immediate, automatic and unconscious reaction to the appearance of an interface. Within a fraction of a second, the brain judges whether a screen looks attractive, calm or chaotic. For older adults, a visceral sense of calm is important. An interface with harsh neon colours or many flashing elements can create an immediate sense of unease, whereas calm colour palettes and generous white space create a sense of calm.

The behavioural level. This level is about functionality, ease of use and physical interaction with the screen. Speed, recognisability and error tolerance determine the emotional outcome at this level. If an older user fills in a form without effort because the buttons are clear and the system immediately confirms each action, the behavioural level produces feelings of competence and control. If the interface fails here, frustration takes over, as shown by Drungilas et al. (2024).

The reflective level. This is the highest level, at which users consciously think about the meaning of the product, their own identity and its long-term value. For older adults, an important reflective question is: "Does this application make me more independent, or does it make me feel old and in need of help?" Interfaces that are designed in a stigmatising way, for example

with childish icons or huge emergency buttons that constantly remind users of their physical vulnerability, fail at the reflective level. A successful age-friendly design integrates accessibility in an attractive, modern way, such as in the principles of Google's Material 3 Expressive, so that older users can feel proud and confident that they can use the system independently.

Approaching interface design through these three levels shows that a usable interface for older adults must do more than meet the functional WCAG guidelines. The design should convey calm at the visceral level, prevent errors and maximise experiences of success at the behavioural level, and support personal autonomy and digital self-esteem at the reflective level.

2.2.4 Design principles that help

The previous sections showed how the visual system works, how it changes in older adults, and which specific barriers result from this. This section answers the logical next question: what can a designer actually do about it? The principles discussed here follow directly from the cognitive and perceptual limitations described earlier. They are not a collection of separate tips but a coherent set of choices that together determine whether an interface is usable.

Visual hierarchy in practice

A screen without a clear hierarchy forces the user to decide what is important. For young, experienced users this takes little effort. For older adults it can be a major obstacle. As described earlier, visual processing slows down with age, contrast sensitivity declines and selective attention becomes less efficient. Hierarchy is the designer's answer to these limitations: by deliberately varying colour, size and position, the design guides the eye before the user actively starts searching.

Colour

Colour is the fastest signal a screen can use to guide the eye. The eye registers colour differences almost reflexively, before shape or position. But colour only works if the contrast is high enough. In older adults, sensitivity to low contrast gradually declines, as described in 2.2.2. A blue label on a dark blue background is already difficult for a twenty-year-old; for someone aged seventy it can be unreadable.

The Web Content Accessibility Guidelines require a minimum contrast ratio of 4.5:1 between text and background. For large text, defined as 18 point or 14 point bold, a minimum of 3:1 applies (W3C, 2018). These thresholds are not arbitrary: they are based on research into the minimum contrast at which text remains reliably readable for people with reduced contrast sensitivity. In practice, these standards are often ignored: light grey text on a white background is one of the most common accessibility errors in modern web design. Figure 3 shows text that is just too light and therefore does not meet the standard for small text.

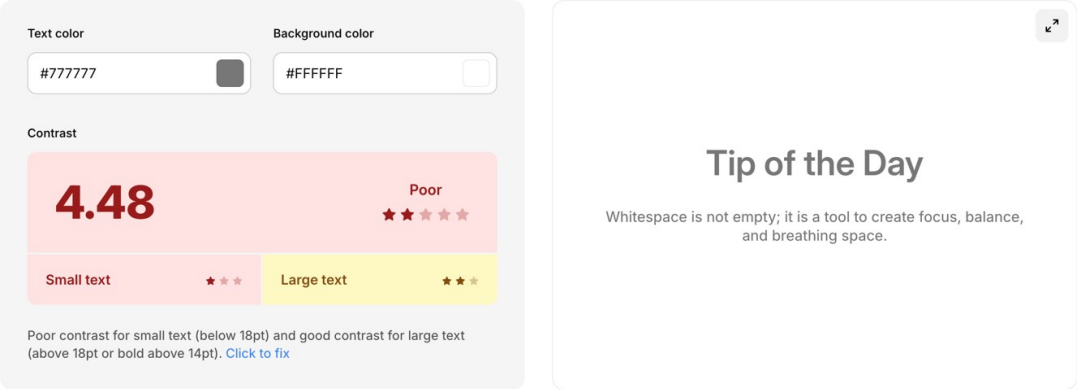


Figure 3. Contrast check in Coolors.co: a monochrome colour combination with a contrast ratio of 4.48:1, just below the standard.

Besides readability, colour also has a hierarchical function. A primary action button in a saturated colour draws the user's eye first. Secondary actions should be less saturated or neutral, so that the user follows the intended order of actions without thinking about it. This only works if colour is used consistently: if every button on the screen has the same colour, the hierarchical signal disappears completely. Figure 4 shows three sets of buttons: in the first, both buttons are saturated; in the second, only one button is saturated; and in the third, neither is. The example shows that contrast guides the eye.

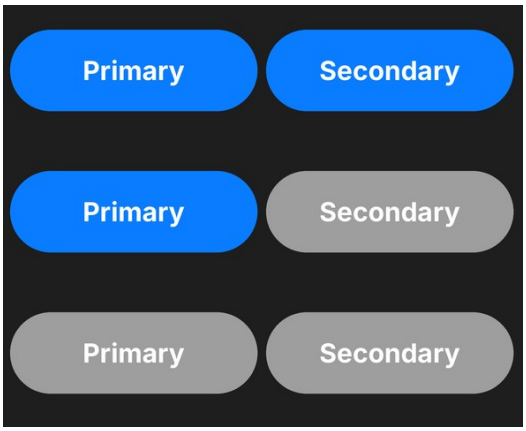


Figure 4. Three pairs of buttons. Only in the middle pair does the primary action stand out from the secondary action.

Colour preference and colour perception in older adults

Colour does not only determine whether a button stands out. It also influences whether an interface feels calm, trustworthy and pleasant. However, there is no colour palette that works for all older adults. Older people differ in eyesight, previous digital experience and cultural associations. Colour preference is therefore a guideline that designers need to test with their own target group, not a fixed rule.

Liu and Wang (2025) studied the colour configuration of smartphone apps for older adults. On average, their participants rated warm hues with high lightness and medium saturation as most pleasant. Dark colour schemes and some cool hues were experienced less positively. This study was carried out in China, so the outcome does not automatically apply to every target group. It does offer a useful starting point: choose a light, calm base and use a clear accent colour that is not garish.

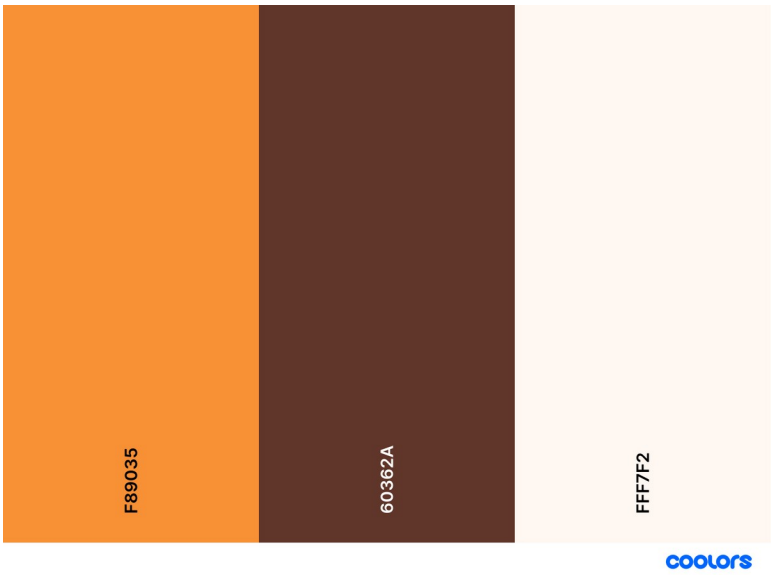


Figure 5. Warm colour palette, made with Coolors.co.

The way colours are perceived also changes with age. Van Leeuwen et al. (2023) found that older adults respond less strongly to differences in saturation, especially for green and magenta. Suzuki et al. (2006) found longer reaction times in older participants for coloured targets on a grey background, particularly for blue and grey; yellowing of the eye lens plays a role here. For an interface, this means that subtle differences, such as light blue versus light grey, are not a reliable way to distinguish information. Dark text on a light background remains more readable than a combination of similar hues.

Moreover, colour should never be the only carrier of meaning. In a study of 80-year-olds, Wijk et al. (1999) showed that colour discrimination and colour preference are not the same: red and yellow were easier to tell apart than blue and green, while blue, green and red scored highest as favourite colours. An error message, a selected option or an important button therefore needs a clear text label, icon or difference in shape in addition to colour. This keeps the interface usable when a user sees a colour less well or simply has a different preference.

Size and typography

Size communicates weight. When one of two elements on a screen is larger than the other, the brain automatically interprets this as a signal of relative priority: the larger element represents the higher level, the smaller one the subordinate level. This is not learned behaviour but a basic

perceptual principle that arises from how the visual system processes stimuli before conscious attention is directed anywhere (Moore & Zirnsak, 2017).

This principle has direct consequences for typographic hierarchy. A heading that is only slightly larger than the body text below it does not send a clear signal. The brain does not register the difference quickly enough to grasp the structure of a page at a glance, which means that the user has to scan longer before it becomes clear where a section begins or what the main question of a form is.

In their heuristic evaluation of supermarket and bank websites, Pascual Almenara et al. (2024) found that an unclear typographic hierarchy consistently emerged as one of the most frequent usability problems, even in interfaces that were otherwise well designed. The problem was not the content but the relationship between the text levels. Figure 6 shows an example of a clear typographic hierarchy.



Figure 6. A clear typographic hierarchy with four clearly different text levels.

Line spacing works in a similar way. When lines are too close together, the eye has to make an effort to find the correct next line every time it jumps back to the left. This increases cognitive load, not because the text is more difficult, but because controlling the eye requires extra work. Line spacing of 1.5 to 1.75 times the font size reduces this by separating the lines enough for the eye to move smoothly (Amouzadeh et al., 2025).

The choice of typeface influences how much effort is needed to recognise individual letters. Sans-serif typefaces such as Arial or Noto Sans reduce the visual complexity of each character, because the fine strokes of serif typefaces are harder to distinguish on screens with a lower pixel density or with declining eyesight. Decorative typefaces add another layer: the brain has to go through more processing steps to recognise the letter shapes before it can process the meaning. This is a form of extraneous cognitive load, the type of load from 2.2.1 that uses working memory without contributing to the task. Figure 7 gives an overview of the types of typefaces.

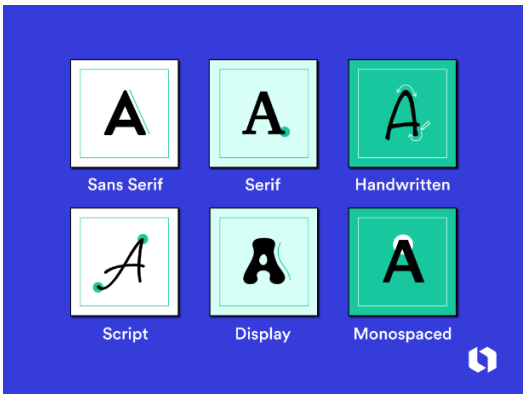


Figure 7. Overview of types of typefaces. Source: Looka (n.d.).

Position and scanning patterns

Position on a screen is not neutral. In Western reading cultures, the eye consistently starts at the top left and then moves horizontally to the right, with attention reaching slightly less far to the right on each following line. This produces the well-known F-pattern that keeps recurring in eye-tracking research. Using questionnaires among 200 users, Qing et al. (2024) confirmed that text and colour are the most influential visual elements on web pages, and that the position of elements strongly guides the scanning order.



Figure 8. Eye-tracking heatmaps showing the F-shaped scanning pattern when reading web pages. Source: Nielsen (2006).

This pattern is explained partly by learned reading habits and partly by the salience route from 2.2.1: at first glance at an unfamiliar screen, the eye goes to what stands out because of position and contrast, and only then does the user search purposefully for what they need (Huang et al., 2023). Elements that are missed in this first phase are less likely to be noticed later.

This has a concrete design consequence: the most critical action or information should be placed where the first glance will catch it. Navigation elements that consistently return in the same position also reduce search time with repeated use, because users build up a spatial memory of where an element is. Any deviation from that expected position breaks this memory and forces the user to search again, which uses working memory that could be better spent on the task itself (Arana-De las Casas et al., 2023).

White space

White space is often treated as an aesthetic matter in design discussions, but it also has a function. According to the Gestalt principle of proximity (see 2.2.1), elements that are close together are read as one group, while larger gaps mark a boundary. White space therefore lets a designer control grouping without the user having to think about it.

White space also lowers the visual density of a screen. The user then has to do less filtering to work out which elements belong together and which action is needed, which saves working memory (Arana-De las Casas et al., 2023).

Consistency and recognisability

Consistency is often treated in the usability literature as one of the basic principles of good design, but the cognitive reason why it works deserves an explanation. With every new screen, users build a mental model: an internal representation of how the interface is organised and what will happen when they tap something. This model is based on previous experience, both within the same application and with other interfaces.

When an interface is consistent in its use of colour, terminology and placement of elements, users can activate and confirm that mental model more quickly. The cognitive load of each new screen drops as a result: the user does not need to learn the structure again but recognises it. Pascual Almenara et al. (2024) identified inconsistency in terminology and visual design as the most frequent weakness in their heuristic evaluations of eight websites, which had an average usability score of only 57.88 percent. This is striking, because these were supermarket and bank websites in active use. Consistency also works across apps. Figure 9 shows iMessage, Instagram, Telegram and WhatsApp: the bars at the top and bottom are almost the same, and all four show message details at the bottom right of each message. Anyone who knows one of these apps will therefore quickly recognise the others.

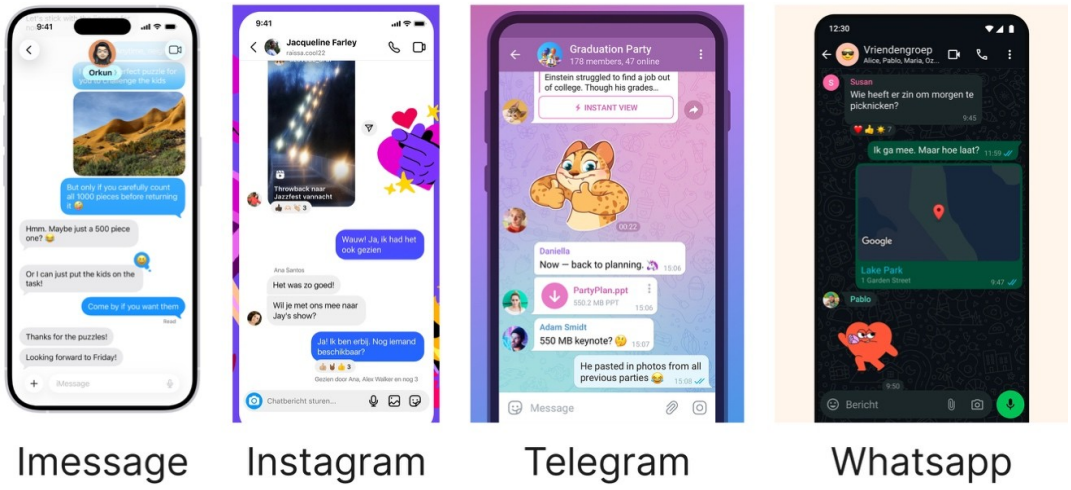


Figure 9. Screens of four widely used messaging apps: iMessage, Instagram, Telegram and WhatsApp.

Dirin et al. (2023) add a dimension that remains underexposed in purely functional usability analyses: consistency is not only a matter of recognition but also of trust. In their model of "feelings of being" for mobile UX, they describe how users develop a sense of security when an interface behaves as expected. If a button on one screen behaves in a certain way and a visually identical button on another screen behaves differently, this sense of security is disturbed. Users start to doubt their own understanding of the interface, which leads to avoidance: tapping unfamiliar elements less readily, not finishing tasks, or leaving the application. This effect occurs in all users, but is stronger in people who already have less confidence in digital environments.

Feedback and error prevention

Feedback is the way a system communicates what it is doing. Without feedback, the user has no information about the state of the system: has my input been received, is the page loading, did the action succeed? This lack of information forces the user to guess, which costs working memory and causes frustration.

Missing feedback is a clear source of extraneous load (see Mental load in 2.2.1): the user spends mental effort on reconstructing the state of the system instead of on the actual task. Drungilas et al. (2024) linked this to measurable emotional reactions: in their framework with integrated emotion recognition, users' expressions of frustration were more strongly related to missing system feedback than to the difficulty of the task. A well-organised interface that leaves users in the dark about what is happening produces as much frustration as a poorly organised interface that does give feedback.

Error prevention works through a related mechanism. When an error has to be corrected on top of several simultaneous decisions, working memory becomes overloaded more quickly (Arana-De las Casas et al., 2023). Error prevention, through clear validation of input fields, confirmation

questions before irreversible actions and avoiding destructive buttons without a warning, reduces the chance of reaching this saturation point. The system takes over part of the monitoring task that would otherwise fall to the user, which frees up working memory for the task itself.

Affordance: intuitive design

The concept of affordance, translated to interface design by Donald Norman from the ecological psychology of James Gibson, describes the extent to which an element communicates how it can be used. An element with a strong affordance shows what can be done with it, without the user having to learn or reason it out. A button that looks like something that can be pressed, a slider that suggests its direction, a text field that shows that you can type in it: all of these reduce the decision load for the user because the interface itself gives the instruction.

The cognitive basis for this lies in what Norman (1988) called perceived affordance: not the objective property of the element, but the user's interpretation of it based on previous experience. This means that affordances are not universal. Wu et al. (2022) demonstrated this in an eye-tracking study in which 54 medical icons were shown to older adults with and without mild cognitive impairment. Flat, minimalist icons, the dominant style in modern interface design, performed worst: older adults spent significantly more time on them and made more search errors when identifying the right function. Skeuomorphic icons combined with a text label performed best. The explanation is that skeuomorphic icons activate existing mental models based on the physical objects they imitate, whereas abstract flat icons require new associations that older adults retrieve less quickly.

Amouzadeh et al. (2025) confirm this in their systematic review of 132 studies: simplified navigation structures and error-tolerant interfaces emerge as the most consistent recommendations. An error-tolerant interface accepts that users will make mistakes and keeps the consequences manageable, for example by offering an undo option. This shifts responsibility from the user to the system, which lowers cognitive load because users need to act less cautiously out of fear of irreversible errors.

2.2.5 Micro-interactions and the perception of time

A micro-interaction is a small, functional moment in an interface in which the system responds to an action or a change of state. Jergović et al. (2024, p. 229) define it as a feedback event that helps complete a task while making the interaction more pleasant, for example pressing a button, turning on a switch or loading a file. Such moments are small, but not irrelevant. According to the same authors, every micro-interaction consists of four parts: a trigger that starts the interaction, rules that determine what happens, feedback that informs the user, and modes or loops that govern the behaviour with repeated use (Jergović et al., 2024, p. 230).

Their importance lies not in aesthetics but in communicating the state of the system. An interface that does not tell users what happens after an action leaves them guessing whether the action succeeded. Jergović et al. (2024, p. 231) tested this with a Figma prototype for disabling a SIM card: 81 percent of respondents found the warning useful, and 74 percent rated the accompanying switch animation as pleasant. In a second prototype, in which users had to

upload a file, the difference between a well-designed and a poorly designed error message directly affected respondents' mood: those who received an unclear error message more often felt frustrated than those who saw a clear success message (Jergović et al., 2024, p. 236).

That micro-interactions are not neutral for older users is shown by the work of Fang et al. (2019). They compared four ways of presenting health information on wearables: text, diagram, static image and animation. The animation interface received the highest usability score on the System Usability Scale from both younger and older respondents, with an average of 80 out of 100 compared with 47.7 for the diagram (Fang et al., 2019, p. 10). Among the older respondents the difference was even larger: an average SUS score of 80.38 for animation versus 52.11 for the diagram (Fang et al., 2019, p. 14). Animation also scored highest on emotional appreciation, with an average of 7.03 on a nine-point scale compared with 4.77 for the diagram (Fang et al., 2019, p. 10). The researchers explain this by noting that animation requires less cognitive interpretation than a diagram full of numbers, while also evoking an emotional response that static interfaces do not (Fang et al., 2019, p. 17).

Button animations and interaction states

An interactive element communicates through change. When a user moves the mouse over a button, presses it or selects it with the keyboard, that difference must be visible before the actual action takes place. These intermediate states, often called hover, active and focus, are micro-interactions themselves. They do not confirm anything definitive, but they do show that the system has noticed the user.

The importance of these states becomes visible when they are missing. Drungilas et al. (2024, §3, Results) built a framework that links users' facial expressions to session recordings in order to detect interface problems that traditional usability tests miss. In one of their test cases, the system registered a cluster of angry emotions at the same time as what Hotjar automatically marked as a "rage click": repeated, rapid clicking on the same element within a minute. The researchers traced this back to an arrow button for which it was not clear whether it had already been pressed, so that users tapped it several times out of uncertainty (Drungilas et al., 2024, §3, Results). The problem was not the function of the button itself, but the absence of a visible change of state.

For keyboard users, this requirement is laid down in a formal standard. WCAG 2.1 Success Criterion 2.4.7 (Focus Visible) requires every operable element to show a visible focus indicator when it is reached with the keyboard (W3C, n.d.-c). The standard gives as examples a vertical bar in a text field showing that you can type, or a visible border around a button that has been reached with the Tab key. This requirement is particularly relevant for people with attention limitations or short-term memory limitations, who benefit from a clear visual reminder of where the interaction currently is (W3C, n.d.-c).

For older users, these two lines of evidence come together. Slow or missing feedback on a button state leads to doubt about whether a tap was registered, and that doubt translates into repeated attempts and frustration, as Drungilas et al. observed. With reduced motor control or

slower reactions, which are more common among older users, the risk of such an unintended double tap is even greater than for younger users.

Loading indicators and the perception of time

Waiting is never neutral. How long a delay feels depends not only on the clock, but also on where the user's attention goes during that delay. Wang et al. (2025, §2.1) build on the Attentional Gate Theory of Zakay and Block: the more attention someone pays to the passing of time itself, the longer that time feels. A loading screen is therefore not merely a technical necessity, but an instrument that can steer the user's attention away from the clock.

Wang et al. (2025) tested this in four experiments comparing a static loading indicator (a ring that does not move) with a dynamic version (the same ring, rotating). In the first experiment, participants who saw the dynamic animation rated the waiting time as significantly shorter than participants who saw the static version, despite an identical objective duration of fifteen seconds (Wang et al., 2025, §4.2.4). A mediation analysis in the second experiment showed that this effect was fully explained by reduced attention to the passing of time, and not by a difference in the visual appeal of the screen (Wang et al., 2025, §4.3.4). Interestingly, the effect depended on context: with high task involvement, such as during a payment process, the difference between dynamic and static animation almost disappeared, whereas it remained strong with low involvement, such as casual browsing (Wang et al., 2025, §4.4.4).

This finding is relevant for designing forms and confirmation screens for older adults: as with the button states above, the absence of movement is not a neutral choice, but one that lengthens the subjective duration of waiting. At the same time, the work of Nontasil and Tangmanee (2024, pp. 336–338) shows that not every form of movement has the same effect. They compared four variants of progress indicators in mobile apps: a bar and a circle, each with and without a ribbed pattern. A progress circle without ribbing reduced perceived delay the most, whereas a progress bar performed better with ribbing (Nontasil & Tangmanee, 2024, p. 341). The researchers conclude that shape and pattern are not independent: which indicator works best depends on the specific combination (Nontasil & Tangmanee, 2024, p. 343).

The timing of the feedback itself turns out to be at least as important as its form. Conrad et al. (2010, p. 417) studied how progress indicators in online questionnaires affect dropout rates, and found that early feedback has a disproportionately large influence on whether someone continues. When the progress indicator suggested a slow pace at the start, 21.8 percent of respondents dropped out; when the same indicator started at a fast pace that then slowed down, this fell to 11.3 percent, with an intermediate rate of 14.4 percent at a constant pace (Conrad et al., 2010). For older users, who according to 2.2.2 are already more sensitive to an unclear system state, this suggests that the beginning of a waiting process matters more than its end.

The Nielsen Norman Group translates findings like these into a practical rule of thumb: for actions that take between two and ten seconds, a simple looped animation is sufficient; for anything longer than ten seconds, a concrete percentage indicator is needed, because otherwise users become impatient with an indicator that does not seem to show real progress

(Sherwin, 2014). As an illustration of what goes wrong without such an indicator, Sherwin (2014) describes a user test in which someone waited patiently for fifteen minutes for a magazine issue to load, without any sign that the system was still active. Exactly that scenario, a spinner that keeps turning without any progress information, is problematic for older users: they cannot tell whether the system is still working or has frozen, and, as with the button states, they already have more difficulty interpreting an unclear system state.

Timing and speed of animations

Not every animation suits every situation, and the speed of an animation changes what it communicates. Yin et al. (2025, §Results) studied three design strategies for the moment just after a tap, when a mobile app loads a new page. They found that a loading icon in a playful, “hedonic” style led not only to a higher perceived loading speed than a plain icon, but also to higher satisfaction. Remarkably, the absence of any icon led to the lowest time perception of the three variants, which suggests that users are less aware of waiting when there is nothing to see at all than when a neutral icon is shown (Yin et al., 2025, §Results). Progressive loading strategies, in which content appears step by step instead of all at once, also reduced the subjective mental load of waiting the most of all variants tested (Yin et al., 2025, §Results).

These findings touch on a tension that appeared earlier in this chapter: an animation that distracts too much is not necessarily better than no animation, and it is the context, not the presence of movement as such, that determines what works. Pibernik et al. (2023, §Results) studied this from a technical perspective and confirm that semantic meaning, meaning an animation that tells the user something about what is happening rather than being purely decorative, shortens perceived waiting time and increases satisfaction. They also set a concrete technical limit: each frame of an animation must be rendered within 16 milliseconds to maintain smooth movement at 60 frames per second (Pibernik et al., 2023, §Results). Below that limit, users experience an animation as jerky, which has the opposite effect of the reassurance a loading indicator is meant to provide.

This technical limit is not an abstraction. Biørn-Hansen et al. (2019, §Results) measured how many frames are actually rendered correctly on different mobile platforms and found large differences: which framework showed the most dropped frames depended on the type of animation, and no single framework performed best in all tests (Biørn-Hansen et al., 2019). For a designer, this means that the choice of technical platform affects whether an animation actually achieves the smoothness needed to be experienced as reassuring rather than disturbing. For older users, who according to Fang et al. (2019) respond more to the emotional tone of an interface than to its information density, a jerky animation is a double risk: it not only lacks the functional reassurance of a smooth animation, but can also create an impression of unreliability that undermines trust in the rest of the interface.

Easing curves in animations

An easing curve determines how the speed of an animation changes over its duration. The horizontal axis shows time and the vertical axis the progress of the movement. With a linear curve, an element moves at a constant speed. An ease-in curve starts slowly and accelerates.

An ease-out curve does the opposite: the element starts moving quickly and slows down before it stops. Ease-in-out combines both, with a gentle start and end.

The benefit of easing is that a transition becomes not only more attractive, but also easier to understand. The change in speed shows that an element is leaving, arriving or changing state. This allows the user to follow the relationship between the old and the new situation. A suitable curve also gives immediate feedback on an action and prevents a transition from feeling like a sudden jump or like a delay in the system. In this way, easing supports visual hierarchy, orientation on the screen and the confidence that the interface responds to input.

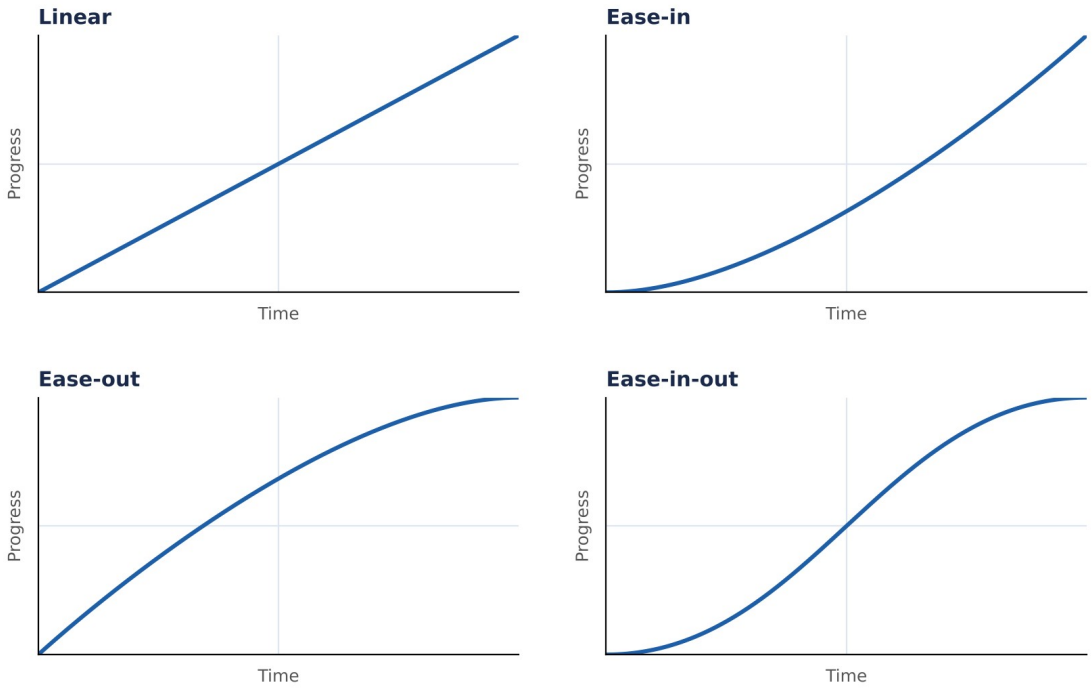


Figure 10. Four commonly used easing curves. The horizontal axis represents time; the vertical axis represents the progress of the animation. Source: own visualisation based on W3C (2023a).

The choice of curve influences how users experience a movement. A linear curve feels constant and predictable, but for a window or button often also mechanical, because real objects usually accelerate and decelerate. An ease-in curve feels hesitant at first and then faster and faster. When a screen opens, that slow start can be experienced as slow feedback, but it suits an element that disappears: it leaves calmly and is then quickly out of view. An ease-out curve feels direct and responsive, because a lot of movement is visible at the start; the gentle deceleration makes clear where the element ends up. Ease-in-out feels balanced and natural because both departure and arrival are gradual. Bouncing, elastic or strongly overshooting curves are often experienced as playful and energetic, but for functional feedback they can distract or suggest that the action has not yet finished. Linear movement does remain suitable for a progress bar, because a constant speed gives an honest picture of the elapsed and remaining time.

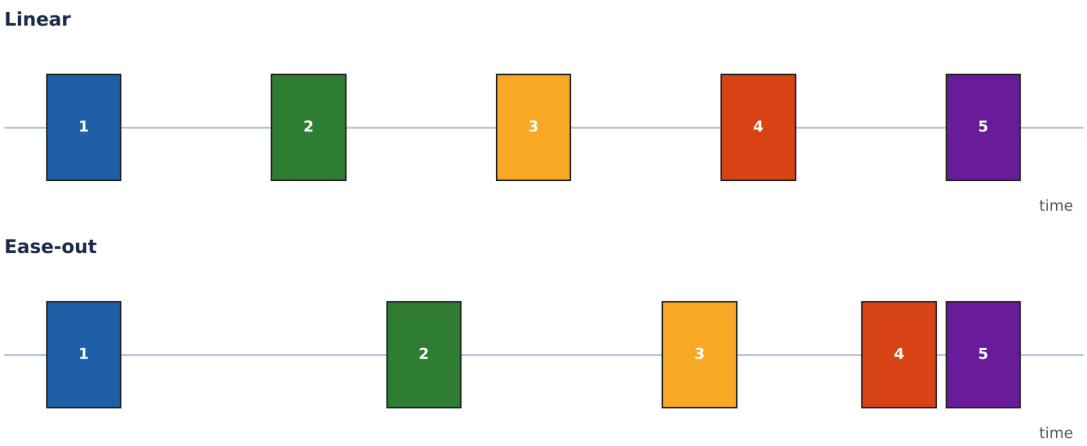


Figure 11. Five moments of the same movement. With ease-out, the element covers more distance at the beginning and comes to rest gently at the end. Source: own visualisation based on W3C (2023a).

The curve may clarify a response of the system, but must not delay it. Ge et al. (2024) found that users experience an interface as less fluent when the delay after a click or the time until loading increases. An animation should therefore start immediately after the action and remain short. Kim et al. (2023) compared, among other things, ease-in and ease-out in screen transitions. Their results show that perceived duration and satisfaction depend on the combination of transition, curve and pause. In their test, a short pause of 150 milliseconds produced more satisfaction than longer waiting times.

For older users, the function of an animation is more important than its decorative effect. Research on motion perception shows that older observers need a larger difference in speed to tell two movements apart, and may have more difficulty perceiving coherence, especially with slow to moderate movement (Snowden & Kavanagh, 2006). The W3C also mentions declining concentration, greater distractibility and limitations in short-term memory among older web users (W3C, 2025a). An abrupt or restless curve can therefore demand extra attention, while a curve with a long, slow start can cause doubt about whether a tap has been registered. For incoming elements, a calm ease-out is therefore often suitable: the response is visible immediately and the deceleration gives time to follow the end position. Ease-in-out suits movements between two visible states, because both start and end are gradual. Strong overshoot, bouncing effects and unnecessarily large movements should be avoided, and the reduced-motion setting of the device should be respected. In this way, the animation supports orientation without competing with text or important information. This is not a fixed rule for every older user: duration and curve should be tested with the target group, and where possible an alternative with little or no movement should be offered.

2.2.6 From theory to design choices

The theory discussed above forms the basis for the prototype we built for this study. We focus on people aged 65 and older with little digital experience, and on two types of task: booking an appointment and finding information. In this section we translate the theory into five design

features: text size, navigation, contrast, micro-interactions and loading feedback. The micro-interactions are part of the base design; the other four features were switched on and off separately in the study. How the prototype works and how we tested the features is described in Chapter 4.

Concrete design guidelines

In their systematic review, Amouzadeh et al. (2025) mention readable text, simplified navigation and sufficient contrast, among other things, as recurring points of attention for mobile interfaces for older adults. The review therefore gives direction to our design choices. However, the applications and user groups studied differ, so that we cannot derive a fixed combination from it that works best for all our participants. We therefore use these guidelines as a starting point for our design variants, not as a proven solution.

Text size

Larger letters can support reading and recognising information, because the ability to see detail and contrast declines with age (see 2.2.2). Yeh (2020) found that larger letters made older adults faster at operating a touchscreen. A font size from that set-up is not a general prescription, however: the screen, the viewing distance and the amount of text also matter. Larger text also comes at a price. The text runs over more lines and the user has to scroll more often (Figure 12). When that happens, the changed layout may also explain why a task goes differently.

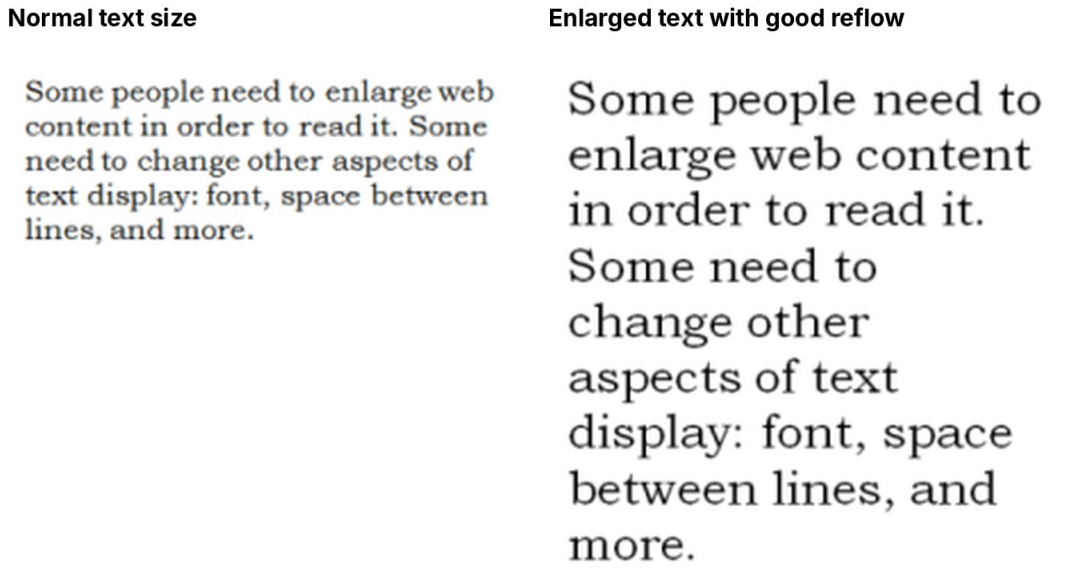


Figure 12. Normal and enlarged text, where the content reflows within the available width. Source: adapted from World Wide Web Consortium (n.d.-b).

Navigation

For navigation, we translate the theory on recognisability, consistency and mental load into visible options, understandable labels and a recognisable way to go back. Options that are visible immediately do not have to be remembered or searched for. Options behind a menu button require an extra action and the knowledge that such a menu exists. Zhou et al. (2022) found indications that layout and information density influence the task performance of older adults, but their study with sixteen participants aged 55 to 76 and a single login task does not prove which navigation works best for our target group and task. Figure 13 shows an example of a fixed, visible navigation bar.

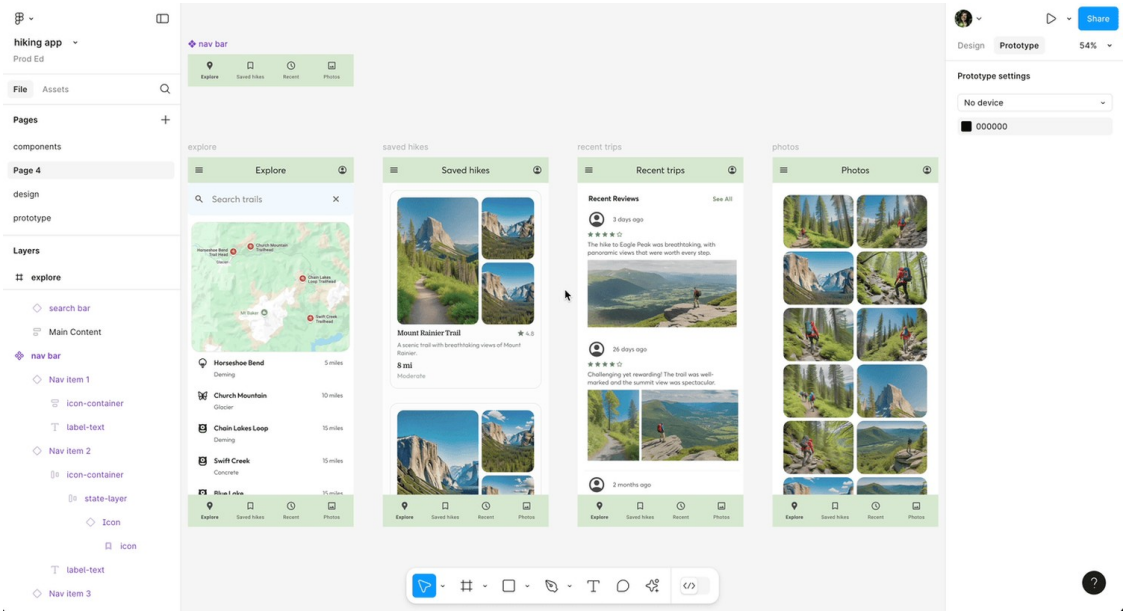


Figure 13. Figma example of a fixed, visible navigation bar that returns on several mobile screens. Source: Figma (n.d.).

Contrast

Contrast concerns the difference between text or controls and their background. The WCAG standards from the section on colour (4.5:1 for normal text and 3:1 for large text) are accessibility criteria, not a guarantee that every older adult can read the text equally easily (W3C, 2023b). Figure 14 shows the same navigation element with insufficient and with sufficient contrast.

The warm colour palette (Figure 5) is separate from the question of how much contrast is needed. Dark brown text on a light cream background stands out clearly, while light beige text on the same background offers little distinction. Our prototype therefore uses the same warm palette at two contrast levels; the exact colour values are given in the method.

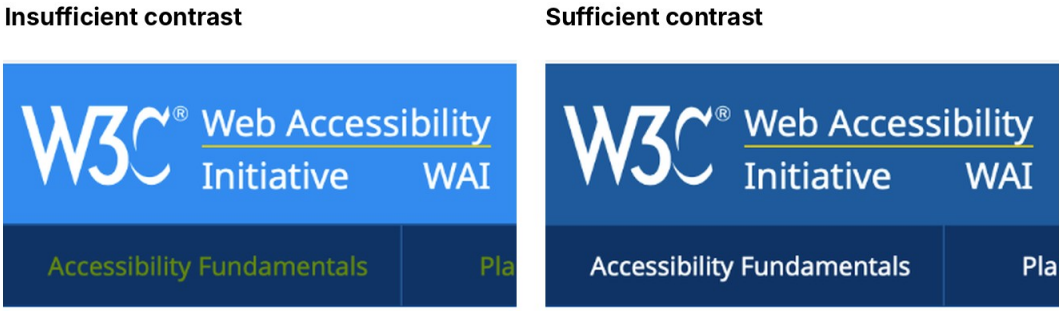


Figure 14. The same navigation element with insufficient and sufficient text contrast. Source: adapted from World Wide Web Consortium (n.d.-a).

Animations

In our prototype, animations have a specific function: showing that a tap has been registered, clarifying a change, or directing attention to relevant information. Section 2.2.5 shows that such micro-interactions can reduce uncertainty about the state of the system, but can also distract if they are too busy or too slow. We therefore use short, calm transitions with an ease-out curve for incoming elements and avoid bouncing or overshooting effects.

Even without movement, it must remain visible that an action has been carried out: a selected option and the final confirmation remain recognisable. Otherwise we would be comparing not only animation, but also the presence of feedback. The W3C also notes that motion can cause discomfort for some users and that non-essential animations should be possible to switch off (W3C, 2025b). The prototype therefore follows the device's reduced-motion setting.

Loading feedback

On real websites, moving between two screens often takes a moment. Section 2.2.5 shows that waiting feels shorter when attention is not focused on the waiting itself (Wang et al., 2025), and that content appearing step by step lowers the mental load of waiting (Yin et al., 2025). A skeleton screen fits these findings: during loading, it already shows the rough layout of the next page as grey blocks. The alternative is a page that stays unchanged during loading, without any sign that the system is busy. Yin et al. (2025) also found that users sometimes notice waiting less when there is nothing to see at all. Whether a skeleton screen helps older adults is therefore not self-evident.

Expressive design and Material 3

Material 3 serves as inspiration for a coherent visual design and recognisable responses to actions. Google describes Material 3 Expressive as a development within its design system in which the emotional experience also plays a role. According to Google, it can be applied in a calm or a more outspoken way (Google, n.d.-a). For our study, the calm application is the most useful: important choices must stand out and responses must be understandable.

In the prototype, this was translated into a clearly recognisable main button and a visible selected state: when someone chooses an option, a marker together with a text label shows which choice is active. Using Material 3 does not in itself prove that a screen is accessible to people aged 65 and older with little digital experience.

Moreover, we do not compare a complete Material 3 design with another design system. The inspiration is used in the shared base of all screens, after which we vary the selected features. This keeps the focus on text size, navigation, contrast and loading feedback, and avoids confusing a general style preference with the effect of a specific feature.

This explains which design features we study and why. Chapter 3 derives the research questions and expectations from this; Chapter 4 describes how we tested the features.

3. Research questions

3.1 Main question and sub-questions

The literature review shows that older adults have more difficulty with small text, low contrast, hidden navigation and an unclear system state, and that such barriers can lead to frustration, uncertainty and eventually avoidance. In our prototype of a fictional healthcare practice, four of these design features can be switched on and off separately: text size, navigation, contrast and loading feedback. The animations from 2.2.6 are part of the base design and were not tested separately. Our main question is:

How can text size, navigation, contrast and loading feedback contribute to a website on which adults aged 65 and older with little digital experience can book an appointment and find information on their phone independently and with as little frustration and uncertainty as possible?

By accessibility we mean here that participants complete a task independently and experience little frustration and uncertainty while doing so. Frustration refers to irritation while carrying out a task; uncertainty refers to doubt about the right next step or about whether an action succeeded. Task duration and errors help to interpret the outcomes. We therefore study a limited part of digital accessibility: the usability of one website on a phone for our participants.

To answer the main question, we look for each feature at what happens when that feature is missing while the other three are switched on. This gives four sub-questions:

1. To what extent does smaller text make it harder to book an appointment and find information independently?
2. To what extent does hiding the next steps behind a menu button lead to more searching, errors and doubt?
3. To what extent does lower contrast between text and background make reading and choosing more difficult?
4. To what extent does the absence of a skeleton screen during an equally long waiting time make participants more uncertain or impatient?

Together, the sub-questions show which feature contributes most for our participants and which one they miss least.

3.2 Hypothesis

We expect that switching off contrast will cause the most problems. Contrast sensitivity declines with age (see 2.2.2), and the W3C (2023b) therefore sets minimum requirements for text contrast. Zhou et al. (2022) also found indications that use of colour and layout are related to older adults' task performance. These sources support the importance of contrast, but do not prove that contrast has the largest effect for our participants; this remains an expectation.

We also expect a clear effect of text size and navigation. Yeh (2020) found that larger letters made older adults faster at operating a touchscreen, so smaller text will more often lead to misreading and doubt. For navigation, we expect that participants without a directly visible next step will search more and get stuck more often, because a hidden menu requires extra knowledge and memory (see Mental load and Affordance). We expect the smallest effect from loading feedback. The waiting time in the prototype is short (1.4 seconds), and Yin et al. (2025) found that users sometimes notice waiting less when there is nothing to see.

4. Method

4.1 Study design

We carried out a small-scale user study with a working web prototype of a fictional physiotherapy practice. The study is mainly qualitative: we looked at what participants did and said, and complemented this with simple quantitative data such as task duration, success and number of errors.

Each participant was given four tasks on a phone, in principle booking an appointment twice and searching for information on the website twice. In each task, exactly one of the four studied features was switched off (text size, navigation, contrast or loading feedback); the other three were switched on. In this way every participant experienced each condition once. There were two exceptions: D02 did three bookings and one information task, and D03 did three tasks, without the contrast condition (see Appendix B). The animations were switched on during all tasks. There was no version in which all features were switched on, and no version in which all were switched off. We therefore compare the four conditions with each other: which feature participants miss most when it is absent.

This differs from our original plan. In that plan, 25 participants would carry out six A/B pairs of bookings on a smartphone or computer, including a separate animation test, a comparison of the complete base design with the fully adapted design, and rating scales after each task. In practice, we reduced the design to four tasks per person, on a phone only, and without the animation test. The main reason was that we found far fewer participants than planned (see 4.2).

4.2 Participants

We looked for participants aged 65 and older, preferably with little experience with websites and apps. First, we asked older people at the market in Valkenswaard to take part. This produced no participants: many older people did not dare to take part because they thought they could not do anything on websites. We then asked grandparents of acquaintances, but only a few of them were available. The group is small and was not selected at random, so the participants do not represent all older adults. Five people took part (D01 to D05), all aged 70 or older. Their experience with smartphones ranged from little or no experience to frequent use. One participant used a stylus and one participant had problems with their eyes and hands. For each participant we recorded the code, the device, the date and notable background information from the conversation; names are not included in the research data.

4.3 Prototype and conditions

The prototype is a web app (React and TypeScript) that was opened in the phone’s browser. To build the website quickly, we used GPT-6 Sol while prototyping. Three participants used an Android phone with Chrome and two used an iPhone with Safari. The website resembles that of a real healthcare practice: a home page with an *Afspraak maken* (Make an appointment) button,

five areas of care, a search bar, a menu, a footer and dozens of information pages. All content, people and the location are fictional; no real appointments or data are stored. Booking an appointment takes four steps (choose a service, choose a date, choose a time, review), followed by a confirmation screen. The calendar shows all working days from October 2026 to March 2027.

For each task, the researcher set the right condition and task in a control panel on the phone, started the participant screen and then handed the phone to the participant. The four features worked as follows; in each condition, only the feature mentioned was switched off.

Text size

When switched on, the site uses 24 pixels for the main text, with larger headings; when switched off, it uses 16 pixels, with smaller headings. Both variants use the typeface Noto Sans with a line height of 1.5. Buttons and option lists are in the same place in both variants.

Navigation

When switched on, the buttons for the next and previous step of the booking route are directly visible on screen. When switched off, the same options are behind a *Menu* button that the participant has to open first. Labels, order and steps are the same.

Contrast

Both variants use a light background (#FFF8F0). When switched on, the text is dark brown (#493B32), well above the WCAG standard of 4.5:1. When switched off, the text is greyish brown (#817971) and borders and buttons are lighter, so that all content apart from images stays below 4.5:1. This low setting exists only as a research condition.

Loading feedback

Each screen transition takes 1.4 seconds in both variants. When switched on, a skeleton screen with the rough layout of the next page and a moving shimmer appears during this waiting time. When switched off, the current page stays unchanged, without any loading signal, until the new page appears.

Animations

In all conditions the micro-interactions were switched on: screens and components appear briefly with a calm movement, and buttons respond visibly to touch. Because the animations were never switched off, we cannot draw conclusions about their effect.

4.4 Tasks

Each session had four tasks: two bookings and two information tasks (for the exceptions, see 4.1). There were four versions of the booking task with a different service, date and time in October 2026, for example “Make an appointment for an intake on Tuesday 6 October 2026 at 10.30”. Each participant was in principle given two of them. The information tasks were: find out

which example location of the practice is mentioned on the site (answer: Valkenswaard centrum), and find the page *Bewegen in een groep* (Exercising in a group) and say who the activity is intended for. In these tasks, participants did not need to make an appointment. Appendix A lists all tasks.

The combination of task and condition and the order of the tasks differed per participant. For example, the low-contrast condition was not combined with the same task for everyone. We did not assign these according to a fixed scheme but chose them per session; the combinations used are listed in Appendix B.

4.5 Procedure

The sessions took place between 8 and 11 October 2026. Each session began with an explanation: we are testing the website, not the participant, and getting stuck says something about the design. Then, for each task, we read out the task and started the participant screen. During the task we gave no hints in principle. When a participant was really stuck, the researcher helped, the task was restarted or the task was stopped; this was recorded. There was no strict time limit. To keep participants unidentifiable, we do not name the locations of the sessions.

After the four tasks, the participant filled in a short online questionnaire with seven open questions: the participant code, which parts were easy, which were difficult or confusing, whether two versions of a task felt clearly different, what helped most when booking the appointments, which feature the participant would like to keep, and any other comments (Appendix A).

4.6 Data collection

We recorded each task with the phone's built-in screen recorder. During the session, the researcher made notes on what stood out and what the participant said. For two tasks the screen recording failed; for these we use only the notes and, where available, a stopwatch time.

After the sessions, we entered the data for each task into a single table: participant code, device, date, order, task, which feature was switched off, result, time, success, the reason for stopping, errors, requests for help, observations and quotes. Time runs from the moment the website appears on screen until the confirmation *Uw afspraak is geboekt* (Your appointment has been booked), or until the answer is on screen. A booking only counts as successful if the service, date and time are all correct. An information task is successful if the participant finds the correct answer. We count as an error a wrong service, date or time and a wrong step in the route; a mistake that the participant corrected themselves does count as an error, but not as a failed task.

We did not ask participants to rate frustration and uncertainty after each task. We derive these experiences from what participants said during and after the tasks, from visible behaviour in the recordings (such as searching for a long time, tapping repeatedly or starting over) and from the answers to the questionnaire. A remark or visible hesitation does not in itself prove frustration;

we therefore describe what we saw and heard and indicate where something is an interpretation.

For some tasks it was not entirely certain afterwards which condition had been switched on, because the control panel was not in the recording. In those cases we derived the condition from what can be seen on screen (for example small text or the *Menu* button) or, if that was not possible, from the conditions the participant had already had. For each task, the dataset records how the condition was established.

4.7 Data analysis

Given the small number of participants, our analysis is descriptive and we do not perform statistical tests. For each condition we count how many tasks were completed successfully without help, how many errors there were and of which type, and we report the median task duration. Because bookings and information tasks differ in nature, we discuss them separately. Times of failed or restarted tasks are not ordinary completion times; we report them separately.

We sort the notes, observations and questionnaire answers into categories, such as reading problems, searching for the next step, unclear feedback, making an appointment by accident and problems with the search bar. We link every observation and quote to the right participant and condition. In this way we can describe for each feature where participants got stuck or hesitated, and which feature they themselves valued most.

We do not make a general ranking of the four features. For each participant, each condition coincided with a different task, and the tasks differ in difficulty. A difference between conditions can therefore also be caused by the task or by the order.

4.8 Validity and reliability

Because every participant experienced all four conditions, individual differences weigh less heavily than they would with four separate groups. To make the measurements verifiable, we recorded every task where possible, and noted for each task how the condition was established and whether the participant was helped or the task was restarted. We have made the prototype, the tasks and the questionnaire public so that others can repeat the test. The limitations of this design, such as the small number of participants, the lack of a version with all features switched on, and inferring frustration and uncertainty without a rating scale, are discussed in sections 7.3 to 7.5.

4.9 Consent and privacy

Before the session, we explained the aim and procedure of the study and asked for consent to take part and to record the screen. Participation was voluntary and participants could stop at any time. The prototype uses only fictional data, so no real appointments or personal data were entered. In the analysis we use codes (D01 to D05). Personal health information that a participant mentioned in the conversation afterwards is included only in summarised form. We do not publish recordings or names. After the research project is completed, we will delete the recordings, except for one that we keep to show what a session looked like.

5. Results

This chapter describes what the five participants (D01 to D05) did during the tasks and what they answered in the questionnaire afterwards. Together they carried out 19 tasks: 11 bookings and 8 information tasks. D03 carried out three tasks, because the contrast condition was missing. One booking (D05, navigation switched off) is not counted for success: the error in the date was partly caused by us, so we deliberately did not assess it. This leaves 18 assessed tasks.

We first give an overview of all tasks (5.1). We then discuss what happened for each feature that was switched off, in the order of the sub-questions (5.2 to 5.5). Section 5.6 presents the answers to the questionnaire and 5.7 discusses how reliable these results are. Because there were only five participants, we always report counts (“1 of 5”) rather than percentages.

5.1 Overview of all tasks

Figure 15 shows for each participant which task was combined with which feature switched off, in which order, whether the task succeeded and how long it took.

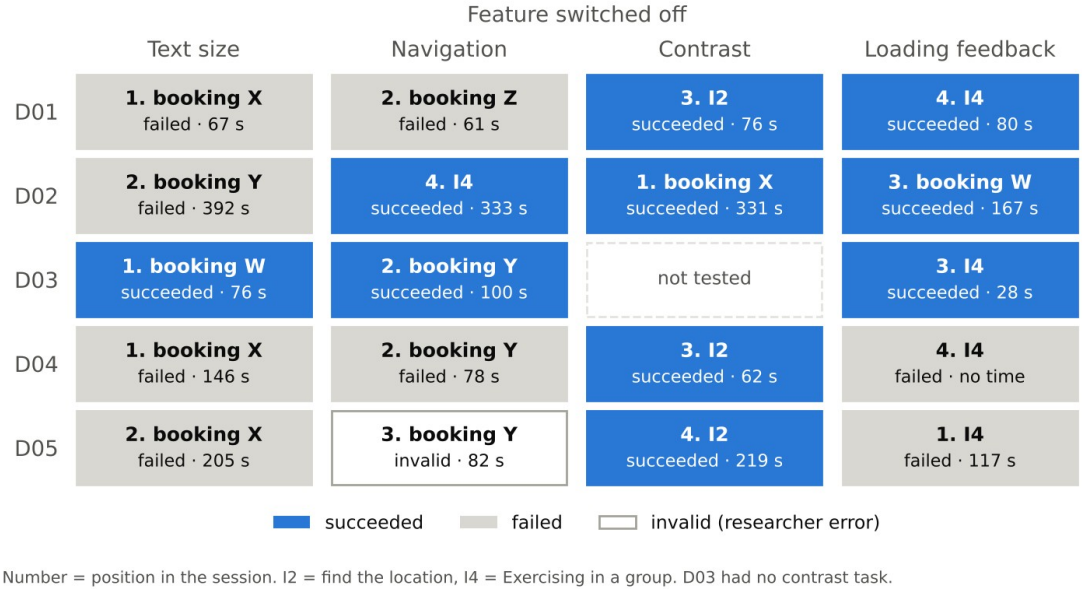


Figure 15. All tasks of the five participants, by feature switched off. The number is the position of the task in the session. I2 = find the example location, I4 = find who Exercising in a group is intended for. For D01-I4 the time was measured with a stopwatch; for D04-I4 no time is available.

Of the 10 assessed bookings, 4 succeeded. Of the 8 information tasks, 6 succeeded. A booking took longer on average than an information task: the median time was 123 seconds for a booking and 80 seconds for an information task.

The figure also shows that feature and task are not independent:

- Text size was switched off during a booking for all five participants, and for four of them this was their first booking.
- Navigation was switched off for four participants during their second booking.
- Contrast was switched off for three of the four participants while finding the location (I2).
- Loading feedback was switched off for four participants during the task about Exercising in a group (I4).

A difference between two features may therefore also be caused by the task or by the position in the session. We return to this in 5.7.

Table 1 and Figure 16 give the number of successful tasks for each feature switched off.

Table 1. Successful tasks and median time of successful tasks, by feature switched off.

Feature switched off	Booking succeeded	Median time (succeeded)	Information task succeeded	Median time (succeeded)
Text size	1 of 5	76 s	–	–
Navigation	1 of 3	100 s	1 of 1	333 s
Contrast	1 of 1	331 s	3 of 3	76 s
Loading feedback	1 of 1	167 s	2 of 4	54 s

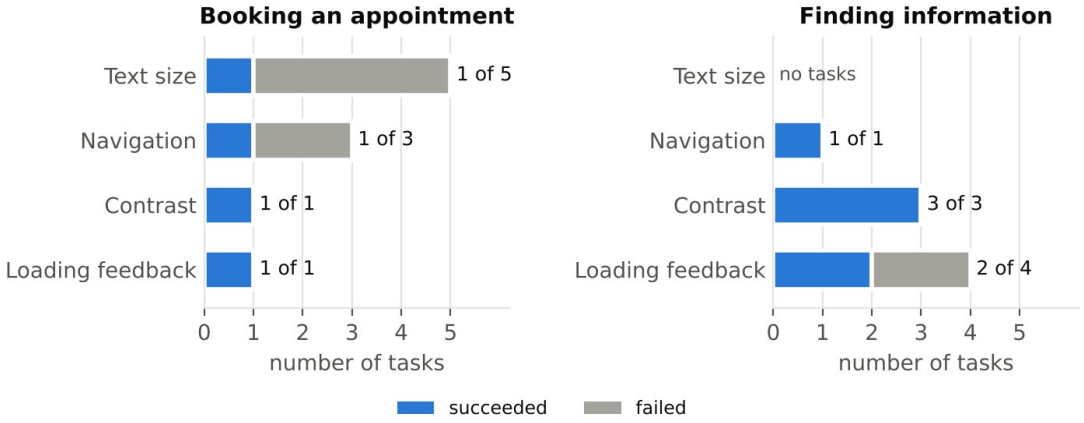


Figure 16. Number of successful and failed tasks by feature switched off, separately for bookings and information tasks. The invalid booking of D05 is not included.

Figure 17 shows the time of each task. Times vary widely, from 28 seconds to more than 6.5 minutes. D02 took longer on every task than the other participants with the same feature switched off (167 to 392 seconds), including the tasks that succeeded. The pace of the participant therefore determines the time more than the feature that was switched off. A failed task was not always slow either: D01 completed the booking without large text in 67 seconds, but with the wrong service and time.

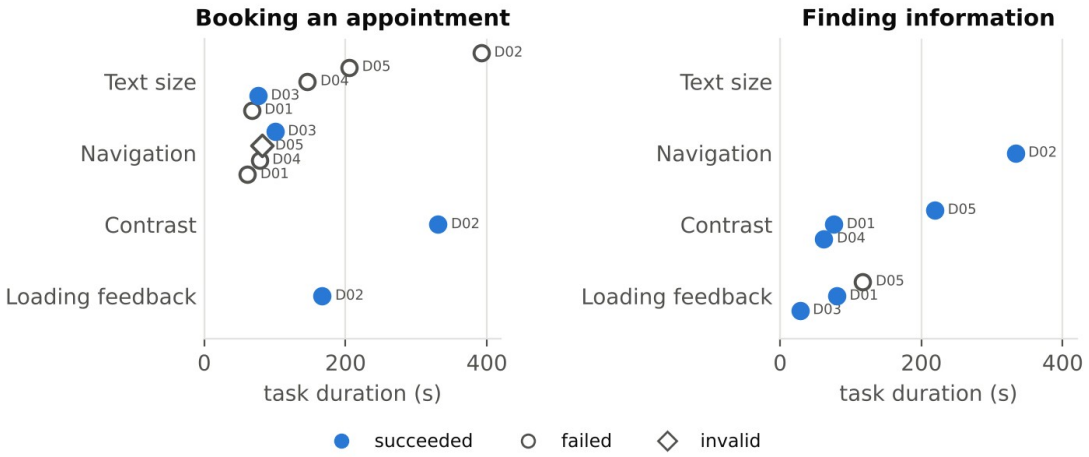


Figure 17. Time per task, by feature switched off. Each dot is one task of the participant named. Tasks without a time (D04-I4) are not shown.

Table 2 gives the types of errors per feature. Errors in service, date and time concern bookings. “Step not found” means that the participant did not know how to continue the booking route. “Other” means, for an information task, that the participant did not find the answer.

Table 2. Types of errors by feature switched off, across all tasks. One task can contain more than one error.

Feature switched off	Tasks	Service	Date	Time	Step not found	Other
Text size	5	3	2	3	0	0
Navigation	5	1	1	1	2	0
Contrast	4	0	0	0	0	0
Loading feedback	5	0	0	0	0	2

5.2 Smaller text

Without large text, 1 of the 5 bookings succeeded. Only D03 booked the correct appointment, quickly and without detours (76 seconds; the recording only started after the *Afspraak maken* button, so the real time is slightly longer).

The other four participants did reach the end of the booking route, but with wrong details:

- D01 booked an appointment within 67 seconds, but chose *Behandeling* (Treatment) instead of *Intake* and a wrong time.
- D02 first searched with the search function, which gave no results, and then via the footer and the *Ergotherapie* (Occupational therapy) page. After 392 seconds, D02 booked an appointment via that page with the wrong service and date.
- D04 first chose *Sportfysiotherapie* (Sports physiotherapy), accidentally left the route, and used the search bar and the menu to get back. On the second attempt, D04 chose *Behandeling* instead of *Intake* and picked a time of their own.

- D05 found the service quickly, but hesitated at every step. D05 could not find the *Verder* (Next) button and tapped below the date. The researcher had to show how to continue and the task was restarted. The appointment was booked with the wrong date and time.

It is striking that the errors were mainly in the chosen service and time (Table 2). These are the choices for which the participant has to read and compare a short text in a list, such as *Intake* and *Behandeling*. We cannot say with certainty that the small text was the cause, but two observations point in that direction. In D04's next task, again with large text, the notes say that the start went more easily because of the large text. And D02 mentioned text size in the questionnaire as the clearest difference between the versions (see 5.6).

5.3 Navigation behind a menu button

Without visible navigation, 1 of the 3 assessed bookings succeeded. In three of the four bookings we saw the participant searching for the next step:

- D01 chose *Oefentherapie* (Exercise therapy) and then searched for 45 seconds between the list of services, the *Menu* button and the overview, without finding *Verder*. The researcher stopped the task after 61 seconds. The notes say: "button not found".
- D03 searched for the next step for about 35 seconds, including in the overview and the footer, and then found *Verder* via the *Menu* button. D03 booked the correct appointment in 100 seconds.
- D04 was briefly confused by the *Menu* button, but continued after about 5 seconds. The recording ends after step 3 without a confirmation, with a wrong service and time filled in.
- D05 had no difficulty with the *Menu* button. This booking is invalid for success (see the introduction to this chapter), but the behaviour is still informative.

D02 had navigation switched off during an information task (I4) and found the answer after a long search via the FAQ (*Veelgestelde vragen*) and the services pages (333 seconds). On information pages, the difference between navigation switched on and off is not visible, so this task says little about navigation.

Navigation was the only feature for which we saw the error "step not found" (twice). In the questionnaire, two participants mentioned navigation: D01 found "finding buttons" difficult and named navigation as a difference between the versions, and D03 found "that menu button in the navigation" the most difficult.

5.4 Lower contrast

All four tasks with low contrast succeeded: finding the location three times and one booking. Still, we saw that reading took more effort for two participants:

- D04 found the location in 62 seconds via the footer. According to the notes, D04 had "slightly more trouble reading", but everything was still visible.
- D05 visibly had more trouble reading. D05 accidentally started making an appointment and, because of the low contrast, did not see that an option had already been selected.

After that appointment, D05 still found the location via the home page and *Praktische informatie* (Practical information) (219 seconds in total).

- D01 found the location in 76 seconds and had not noticed the low contrast.
- D02 booked the correct appointment, but took a long time (331 seconds). D02 tapped for the first time only after more than a minute and scrolled through the list of services for a long time. This was D02's first task, and D02 was slow in all tasks.

In the questionnaire, D02 mentioned contrast, alongside text size, as a clear difference: according to D02, the text should be black. D02 explained this by referring to their own problems with their eyes and hands.

5.5 No loading feedback

Without a skeleton screen, 3 of the 5 tasks succeeded: D02's booking and two of the four information tasks. In the recordings and notes we saw no behaviour related to waiting, such as tapping repeatedly or asking whether the page was loading. According to the notes, D01 had "no problems without a loading indicator". No participant mentioned loading in the questionnaire.

The two failed tasks were both the task about Exercising in a group, and they failed in the same way: the participant made an appointment instead of looking up the answer.

- D04 reached the page via the footer, thought the target group was "Alex" (a name from the booking form) and eventually made an appointment.
- D05 reached the correct overview page after 47 seconds, but tapped *Afspraak maken* there and booked an appointment. D05 needed some help to understand that an appointment was not necessary. The search bar still contained the word "Afspraak" (appointment) from an earlier search, and D05 thought the search button led to a separate search page.

These errors are related to the task and to the *Afspraak maken* button, not to loading (see also 5.7).

5.6 Answers to the questionnaire

All five participants filled in the questionnaire after the four tasks. Table 3 summarises which features they mentioned.

Table 3. Features that participants mentioned in the questionnaire.

Question	Text size	Navigation	Contrast	Loading feedback
Difficult or confusing	–	D01, D03	–	–
Clear difference between versions	D02	D01, D03	D02	–
Would like to keep	D02	D03 ("remove menu")	–	–

Navigation was mentioned most often: by two participants as difficult and as a clear difference. Text size and contrast were each mentioned by one participant. Nobody mentioned loading

feedback. D04 noticed no difference between the versions, and D04 and D05 did not know which feature they would like to keep.

Other answers were about the type of task. D05 found making an appointment and choosing a date easy, but “finding things on the site difficult”, and found the last task the most difficult: “with the other tasks you could tap straight away, the rest you really had to search for”. D03 found choosing the date easy. D02 wrote that it was not confusing, but “just a matter of not being used to this”. D04 had “more trouble with the questionnaire than with the site”. As help with the appointments, participants mainly mentioned looking carefully and the task card (“with the cheat sheet for the appointment”, D02).

In the conversation after the test, D05 suggested three improvements: clear the search bar every time, add a note that you can search there, and also place the *Verder* button below the date. D05 also said that their own phone always involves too much searching and that they always make appointments by phone call.

Apart from the four features, we saw two problems that recurred with several participants:

- **Making an appointment by accident.** In 3 of the 8 information tasks (D04-I4, D05-I4 and D05-I2), the participant started making an appointment instead of looking for information.
- **The search bar.** D02 got no search results, D04 used the search bar to get back into the booking route, and for D05 an old search term remained in the bar.

5.7 Evaluation of the results

Four points are important when reading these results.

The number of participants is small. For each feature there are four or five tasks, divided over two types of task. For some combinations there is only one task (such as the booking with low contrast). One participant more or less could therefore change the picture completely. The counts are a first indication, not proof.

Feature, task and order coincide. As Figure 15 shows, each feature was mainly combined with one type of task:

- Text size almost always coincided with the first booking, when the participant did not yet know the route. The poor score for text size (1 of 5) may therefore also be a learning effect: in the second booking, participants already knew the steps.
- Contrast mainly coincided with finding the location. This task always succeeded and is probably the easiest.
- Loading feedback mainly coincided with the task about Exercising in a group. According to D05 this was the most difficult task, and both failures there were caused by the *Afspraak maken* button.

As a result, the success counts per feature mainly reflect the difficulty of the task. We therefore cannot derive a ranking of the features from Table 1.

Behaviour says more than the counts. The observations that fit what a feature does are stronger. Without visible navigation, three of the four participants searched for the *Verder* button, and the error “step not found” only occurred with navigation. Two participants named the menu button itself as difficult. With small text, the errors were mainly about choosing the right service and time from a list, and two participants mentioned text size themselves. With low contrast all tasks succeeded, but two participants visibly had more trouble reading. Without loading feedback, we saw and heard nothing that pointed to uncertainty caused by waiting. This fits the short waiting time of 1.4 seconds.

The times are of limited use. The differences in time are mainly caused by the participant (D02 was the slowest for every feature), not by the feature. In addition, several times are not fully comparable:

- For D03-W, the recording only started after the *Afspraak maken* button.
- For D01-I4, the time was measured with a stopwatch.
- For D04-I4, no time is available.
- For D05-X, the restart is included in the time.
- For D02-W, time was measured from the restart, after a false start.

We did not measure frustration and uncertainty with a rating scale. What we say about them, such as hesitating, searching for a long time or needing help, is what we saw in the recordings and notes, and remains an interpretation.

6. Conclusion

In this chapter we first answer the four sub-questions (6.1 to 6.4) and then the main question (6.5). In 6.6 we look back at our hypothesis. In 6.7 we list what our design solves according to the results and where it still falls short. In 6.8 we indicate how robust these conclusions are. We base ourselves only on the five participants from Chapter 5.

6.1 Smaller text

The first sub-question was to what extent smaller text makes it harder to book an appointment and find information independently.

Smaller text probably made booking an appointment more difficult for our participants. Without large text, 1 of the 5 bookings succeeded. The other four participants did reach the end of the booking route, but with a wrong service, date or time. The errors were mainly in the service and the time: choices for which the participant has to read and compare short texts in a list. Two participants indicated themselves that text size made a difference.

We cannot separate the effect of text size from the order. For four of the five participants, the small text coincided with their first booking, when they did not yet know the route. Some of the errors may therefore also be due to the participant seeing the route for the first time. This study says nothing about finding information, because text size was not switched off in any information task.

6.2 Navigation behind a menu button

The second sub-question was to what extent hiding the next steps behind a menu button leads to more searching, errors and doubt.

Navigation shows the clearest effects. Without visible buttons, three of the four participants searched for the next step in the booking route. One of them did not find the *Verder* button at all, after which the researcher stopped the task. The error “step not found” only occurred with this feature. Of the three assessed bookings, one succeeded. Two participants named the menu button in the questionnaire as difficult. For our participants, hiding the next steps therefore led to more searching and more errors, and two of them searched for 35 to 45 seconds without knowing how to continue. Not everyone was affected: one participant used the menu button without difficulty.

We cannot draw conclusions about information tasks for navigation. On the information pages, the difference between navigation switched on and off is not visible.

6.3 Lower contrast

The third sub-question was to what extent lower contrast between text and background makes reading and choosing more difficult.

Lower contrast made reading more difficult for some participants, but did not stop anyone. All four tasks with low contrast succeeded. Two participants did visibly have more trouble reading, and one of them did not see, because of the low contrast, that an option had already been selected. One participant mentioned contrast as a clear difference; another had not noticed the low contrast. For our participants, lower contrast therefore mainly affected reading and recognising a selected option, not task success.

This conclusion is weak. For three of the four participants, the low contrast coincided with finding the location, probably the easiest task. There was only one booking with low contrast.

6.4 No loading feedback

The fourth sub-question was to what extent the absence of a skeleton screen during an equally long waiting time makes participants more uncertain or impatient.

We found no indication that the absence of loading feedback made our participants more uncertain or impatient. In the recordings and notes we saw no behaviour related to waiting, and no participant mentioned loading. The two tasks that failed without loading feedback failed because the participant made an appointment instead of looking for information, not because of loading. With a waiting time of 1.4 seconds, loading feedback therefore made no noticeable difference for our participants. We did not study whether this also holds for longer waiting times.

6.5 Answer to the main question

Our main question was: *How can text size, navigation, contrast and loading feedback contribute to a website on which adults aged 65 and older with little digital experience can book an appointment and find information on their phone independently and with as little frustration and uncertainty as possible?*

For our five participants, visible navigation and large text in particular contributed to booking an appointment independently. Without visible buttons for the next step, participants searched for the next step and sometimes got stuck. Without large text, they more often chose the wrong service or time. High contrast made reading easier, but its absence did not stop anyone. Loading feedback made no noticeable contribution with a short waiting time. For finding information, we saw no clear difference between the features.

Together, the four features are not enough to make independent booking certain. In every task, three of the four features were switched on, and still only 4 of the 10 assessed bookings succeeded. Of the information tasks, 6 of 8 succeeded. Many of the problems had nothing to do with the four features: participants made an appointment by accident, were confused by the search bar, or could not find the *Verder* button when it was out of view. Whether an older adult can book an appointment independently therefore depends not only on text size, navigation, contrast and loading feedback, but also on how the tasks on the website are organised.

We did not measure frustration and uncertainty with a rating scale. What we saw was hesitation, long searching and sometimes needing help. We saw this mainly with hidden navigation, with small text and in the task about Exercising in a group.

6.6 Looking back at the hypothesis

Our hypothesis was that switching off contrast would cause the most problems, that text size and navigation would also have a clear effect, and that loading feedback would have the smallest effect.

- **Contrast:** not confirmed. With low contrast all tasks succeeded, although reading took more effort for two participants. Because contrast mainly coincided with the easiest task, the hypothesis is not refuted either: this study could not measure the difference well.
- **Text size and navigation:** largely confirmed. For navigation we saw the expected searching and getting stuck most clearly. For text size we saw the expected errors in choosing, but these may partly be due to a learning effect.
- **Loading feedback:** confirmed. We saw no effect at all of the absence of loading feedback.

6.7 What our design solves and what could be better

Because there was no version in which all features were switched on, we cannot say how much our complete design helps compared with an ordinary website. The results do show which parts of the design do their job:

- **Visible next-step buttons** prevent participants from having to search for the next step. This is the part for which we have the most evidence.
- **Large text** is associated with fewer errors when choosing a service and a time.
- **High contrast** makes reading and seeing a selected option easier for participants with poorer eyesight.

The results also show where the design falls short:

- **The *Afspraak maken* button attracts too much attention.** In 3 of the 8 information tasks, the participant started making an appointment instead of looking for information. The prominent main button, which we deliberately made easy to recognise, also attracted attention when it was not needed.
- **The search bar does not work well enough.** One search gave no results, an old search term remained in the bar, and one participant used the search bar to get back into the booking route.
- **The *Verder* button is not always in view.** One participant looked for the button below the date and needed help, and choosing a date required further scrolling.
- **Services are hard to tell apart.** Participants chose the wrong service four times, for example *Behandeling* instead of *Intake*. This happened mainly with small text, but once also with large text.

The results therefore suggest these improvements: a *Verder* button fixed at the bottom of the screen, a search bar that always starts empty and shows that you can search there, a less dominant booking button on information pages, and a short explanation for each service. Some of these were suggested by the participants themselves.

6.8 Validity of the conclusions

These conclusions apply to our five participants, this website and these tasks on a phone. They are first indications, not proof, for three reasons:

- Five participants took part, and they were not selected at random.
- Each feature mainly coincided with one type of task and a fixed position in the session, so that we cannot properly separate the effect of a feature from the task and from a learning effect.
- We inferred frustration and uncertainty from behaviour and remarks, and did not measure them with a rating scale.

The conclusion about navigation is the strongest, because the behaviour in the recordings fits what the feature does and participants mentioned it themselves. The conclusions about contrast and loading feedback are the weakest, because they rest on few tasks of one type. We cannot give a ranking of the four features. Chapter 7 discusses these limitations further.

7. Discussion

In Chapter 6 we answered the sub-questions and the main question and described what our design solves and what could be better. In this chapter we look critically at the study itself. Section 7.1 discusses why the results do or do not match our expectations. Section 7.2 covers problems with the prototype as a research instrument. Sections 7.3 to 7.5 discuss the reliability, validity and representativeness of the study, and what we did to limit these problems. Section 7.6 considers what the results might mean, and 7.7 explains why we made the study public and gives suggestions for further research.

7.1 Results compared with expectations

Navigation caused the problems we expected. A menu button requires an extra action and the knowledge that something is hidden behind it (see Mental load in 2.2.1 and Affordance in 2.2.4). Amouzadeh et al. (2025) name hidden functions and complicated menus as recurring problems for older adults. We saw this too: participants searched for the next step and one participant did not find it at all. Remarkably, the amount of experience did not predict who had trouble with the menu button. D05, with little or no smartphone experience, used the menu button without difficulty, while D01 could not find the *Verder* button. At that point, however, D05 had already gone through the booking route once and had been shown the *Verder* button. This may therefore also be a learning effect.

Text size gave roughly what we expected: smaller text went together with more errors in choosing, as Yeh (2020) also found for older adults and touchscreens. It is important to note that our small text was not extremely small. The switched-off setting used 16 pixels, the default size in most browsers and on many ordinary websites. If the difference really is caused by text size, this means that an ordinary website can already cause difficulties when users have to choose from a list of short, similar options. As noted in 6.1, part of the difference may also be due to a learning effect.

Contrast caused fewer problems than we expected. We think there are three reasons for this. First, the low contrast was fairly mild. The greyish brown text (#817971) on the light background (#FFF8F0) has a contrast ratio of about 4.1:1, just below the standard of 4.5:1 for normal text. For large text, WCAG only requires 3:1 (W3C, 2023b). Because the large text of 24 pixels was switched on in the contrast condition, the main text in that condition still met the standard for large text. Second, large text and high contrast may compensate for each other: larger letters remain readable even with less contrast. Because only one feature was switched off at a time, we could not observe this. Third, the low contrast mainly coincided with the easiest task, finding the location. The only participant who mentioned contrast in the questionnaire had problems with their own eyes. This fits the decline in contrast sensitivity described in 2.2.2: contrast seems to matter mainly for people with poorer eyesight.

Loading feedback had, as expected, no noticeable effect. This fits the literature. The waiting time of 1.4 seconds is below the two seconds from which Sherwin (2014) recommends a loading animation. Yin et al. (2025) found that users sometimes notice waiting less when there

is nothing to see. And Wang et al. (2025) found that the difference between loading animations almost disappeared when people were concentrating on a task, such as during a payment. Our participants were always working on a clear task. The consequence is that our design could only partly answer the fourth sub-question: with a short waiting time, there was little uncertainty to reduce.

What we did not expect was how many problems had nothing to do with the four features. The prominent *Afspraak maken* button, which we had deliberately made easy to recognise (see 2.2.6), drew participants away from their goal during information tasks. This shows that a choice that helps in one task can work against the user in another. Lam (2024) points out that guidelines written for people with disabilities do not always give older adults a good experience. Our results are in line with this: with three of the four features switched on, only 4 of the 10 bookings succeeded.

7.2 Problems with the prototype as a research instrument

Section 6.7 described what could be improved in the design of the website. This section is about something else: problems in the prototype that affected the measurements themselves.

- **The navigation condition only worked in the booking route.** On information pages, the site looked the same with navigation switched on and off. D02's information task with navigation switched off therefore did not really test a condition.
- **The condition was not always visible in the recording.** The control panel was not on screen for every task. For several tasks we therefore derived the condition from what is visible on screen, and for two tasks only by elimination.
- **The search bar caused problems unrelated to the condition.** One search gave no results and an old search term remained in the bar. This affected tasks in different conditions.
- **The *Verder* button was not always in view.** When choosing a date, the participant had to scroll further. Because of this, one booking of D05 is invalid. D05 also failed to find the *Verder* button with navigation switched on. Searching for the next step is therefore not caused only by the menu button, although we saw it most often there.
- **There were technical problems.** During one of D02's tasks, the photos on the site did not load, and two screen recordings failed.

Many of these problems only came to light during the real sessions. As a result, some of the failed tasks were caused by the base design rather than by the feature that was switched off. In Chapter 5 we indicated this for each task as well as we could.

7.3 Reliability

Reliability concerns whether we would obtain the same results if the study were repeated. We are cautious about this. With five participants, one participant more or less can completely change the picture for a feature. For some combinations of feature and task there is only one observation. Task duration depended mainly on the pace of the participant: D02 was the slowest for every feature.

The procedure was also not the same in every session. We chose the combination of task, condition and order per session rather than following a fixed scheme. When a participant got stuck, the researcher helped, the task was restarted or the task was stopped. Which of the three happened depended on the situation. As a result, some tasks are hard to compare with each other. The starting point of the time measurement also differs a few times, because a recording started too late or was missing.

To limit these problems, we did three things. We recorded every task (apart from the two failed recordings), so that the times and errors can be checked in the video. The dataset records for each task how we know the condition and the task, and whether the participant was helped or the task was restarted. And we did not count the booking that went wrong partly because of our own mistake. The recordings were, however, analysed only by us. A second assessor who rates the same recordings independently would increase reliability.

7.4 Validity

Validity concerns whether we measure what we intend to measure. This is limited in three respects.

We did not measure frustration and uncertainty directly. In our original plan, participants would give a rating after each task. In practice this was dropped, and we inferred frustration and uncertainty from behaviour, remarks and the questionnaire afterwards. Visible hesitation or long searching may also mean that someone is being careful. Moreover, we designed the website ourselves and knew which outcome we expected. To limit this, we described in Chapter 5 what we saw and indicated separately where we were interpreting.

The design measures what participants miss, not what the design achieves. In every task, exactly one feature was switched off and the other three were switched on. There was no version with everything switched on and no version with everything switched off. We therefore cannot say how much our design helps compared with an ordinary website. Features that compensate for each other, such as large text and contrast, may also appear less important in this design than they really are. In addition, each feature mainly belonged to one type of task and a fixed position in the session (see 5.7). We therefore often cannot separate whether a difference is caused by the feature or by the task.

The tasks only partly resemble real life. Participants received a task card with the service, the date and the time. They mainly had to find and copy these details, not work out themselves what they needed. One participant even called the card the “cheat sheet” that helped most. In real life, someone may not know whether they need an intake or a treatment, which makes choosing a service even harder. In addition, a researcher was always sitting next to the participant and could help. At home, nobody tells you that an appointment is not necessary.

7.5 Representativeness

Our five participants are not representative of all older adults. They were not selected at random. We first asked older people at the market in Valkenswaard to take part, but the older people we spoke to there did not dare to take part because they thought they could not do

anything on a website. We then asked grandparents from our own circle of acquaintances, of whom only a few were available. Our participants are therefore people whom the researchers or their families know, and who dared to take part. All participants were 70 or older; for privacy reasons we do not report more precise ages. The group was nevertheless varied: smartphone experience ranged from little or no experience to frequent use, one participant used a stylus and one participant had problems with their eyes and hands. Not all participants therefore belonged to the target group “little digital experience” from our main question. Jiang et al. (2024) point out that older adults differ greatly in digital experience. Five people are too few to cover these differences.

The older adults we most wanted to reach therefore did not take part. At the market we saw that older people who think they cannot do anything on a website also do not want to take part in a test. For them, an accessible website may matter most. D05 told us that they always make appointments by phone call, because their own phone involves too much searching. This shows that the digital divide described by Mubarak and Suomi (2022) does not disappear through better design alone.

In addition, the results apply to one fictional physiotherapy website, on a phone, with three Android phones and two iPhones. This study says nothing about computers, tablets, screen readers or long-term use. We therefore cannot draw general conclusions about older adults and websites from this study. Our results are first indications that need to be tested with more participants.

7.6 What the results might mean

With these limitations in mind, we see three possible implications.

First, a visible next step seems at least as important for older adults as adjustments to text and colour. Many mobile websites hide their navigation behind a menu button to save space. For our participants, that was exactly where they most often got stuck. If this is confirmed with more participants, it is a simple and cheap improvement.

Second, a normal text size may not be enough in places where older adults have to choose between short, similar options, such as services and times. Our participants made the most errors in those places.

Third, accessibility for older adults is about more than text, colour and buttons. For our participants, much went wrong because of how the tasks were organised: a button that attracts attention when it is not needed, a search bar that does not do what someone expects, or a button that is out of view. Designing a website for older adults therefore also means thinking about what someone wants to do on each page, and getting everything else out of the way.

7.7 Open research and further research

Because we had only five participants, our results can be no more than a first indication. We have therefore made the study public, so that others can repeat it with more participants. The repository contains the prototype with the controls for each condition, the tasks, a step-by-step

guide for running a session, and blank forms for recording results and obtaining consent. [TO FILL IN: link to the repository.] Anyone who carries out the study with the same tasks and recording forms can compare their results with ours. In this way, small studies can together give a better picture than one study alone.

For further research we recommend:

- **More participants**, recruited through different channels, with age and digital experience recorded in advance.
- **A fixed scheme** in which each condition occurs equally often with each task and at each position in the session. Feature, task and learning effect can then be separated.
- **A version with all features switched on and a version with everything switched off**, so that it can be measured how much the complete design helps and whether features compensate for each other.
- **A short rating scale after each task** for frustration and uncertainty, in addition to the observations.
- **A longer waiting time** for the loading feedback, for example between two and ten seconds, and a separate test of the animations, which were always switched on in our study.
- **A pilot session first**, in which the problems from 7.2 are found and solved, and a prototype that records the condition itself.
- **An improved version of the website** with the changes from 6.7, tested against the current version.
- **A second assessor** who analyses the recordings independently of the researchers.

8. Afterword

[TO FILL IN by Yannis and Billie: reflection on the process and the collaboration.]

This English version was translated from our Dutch report with the help of an AI tool and checked by us.

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Appendix A. Tasks and questionnaire

The researcher read out each task in Dutch. The booking tasks always had the form “Make an appointment for [service] on [day and date] at [time].”

- Booking X: intake (*Intake*) on Tuesday 6 October 2026 at 10.30.
- Booking Y: treatment (*Behandeling*) on Thursday 8 October 2026 at 13.00.
- Booking Z: exercise therapy (*Oefentherapie*) on Monday 19 October 2026 at 09.00.
- Booking W: manual therapy (*Manuele therapie*) on Wednesday 28 October 2026 at 14.30.
- I2: “Find out which example location of the practice is mentioned on the site.” Expected answer: Valkenswaard centrum.
- I4: “Find the page *Bewegen in een groep* (Exercising in a group) and say who the activity is intended for.” Expected answer: people who want to practise or exercise together with others.

After the four tasks, participants filled in the online questionnaire “PWS – Vragen na de hele test” (Questions after the whole test), with the following open questions (translated from Dutch):

1. What is your participant code?
2. Which parts did you find easy?
3. Which parts did you find difficult or confusing?
4. Did two versions of a task feel clearly different? Which was easier or more difficult, and why?
5. What helped you most when making the appointments?
6. Which feature would you like to keep?
7. Is there anything else you would like to say?

Appendix B. Overview of the tasks carried out

Table B1. Tasks per participant, with the feature switched off and the outcome.

Participant	Order	Task	Feature off	Condition established by	Succeeded	Time (s)
D01	1	Booking X	text size	appearance in recording	no	67
D01	2	Booking Z	navigation	panel on screen	no	61
D01	3	I2 location	contrast	appearance in recording	yes	76
D01	4	I4 Exercising in a group	loading feedback	inferred	yes	80*
D02	1	Booking X	contrast	panel on screen	yes	331
D02	2	Booking Y	text size	panel on screen	no	392
D02	3	Booking W	loading feedback	panel on screen	yes	167
D02	4	I4 Exercising in a group	navigation	inferred	yes	333
D03	1	Booking W	text size	appearance in recording	yes	76
D03	2	Booking Y	navigation	appearance in recording	yes	100
D03	3	I4 Exercising in a group	loading feedback	appearance in recording	yes	28
D04	1	Booking X	text size	appearance in recording	no	146
D04	2	Booking Y	navigation	panel on screen	no	78
D04	3	I2 location	contrast	appearance in recording	yes	62
D04	4	I4 Exercising in a group	loading feedback	inferred	no	–
D05	1	I4 Exercising in a group	loading feedback	appearance in recording	no	117
D05	2	Booking X	text size	panel on screen	no	205
D05	3	Booking Y	navigation	appearance in recording	not assessed	82
D05	4	I2 location	contrast	appearance in recording	yes	219

Note. A booking only counts as successful if the service, date and time are all correct. Times marked with * were measured with a stopwatch because the recording is missing. For D04 (task 4) the recording failed and no time is available. D05 (task 3) is not counted for success,

because the date error was partly caused by the researchers. D03 did not carry out a task with contrast switched off.